

Group symbol: **2**

Team: **1**

Project title: **ListItUp**

Team members (*filled by PM, Team Leader*):

No	Name	Surname	Student ID	Role
1	Carmelo	Romero Pérez	293936	<i>PM, Team Leader</i>
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4. Construction and tests (S4)

4.1. Implementation

The documentation focuses on the implementation of the software in the form of programming language source codes. You should prepare lists of created artefacts, i.e., source codes, configuration files, files containing program resources, pre-requirements, and result programs. You should supplement the list of artefacts with a description indicating the relationships between them and their nature (type), as well as technical requirements in the scope of their use. The implementation should cover the minimum viable product (MVP) of the system.

The presentation of the application execution is a must for obtaining a positive grade.

This section documents the software implementation of ListItUp, covering source code, configuration files, resource files, and result artefacts. It describes how the deliverables relate to technical and functional requirements, and confirms the scope of the Minimum Viable Product (MVP). A live deployment of the application is accessible at <https://listitup.duckdns.org>

4.1.1. Overview

ListItUp is a web-based social list-curation platform that allows users to create, share, and discover ranked lists about any topic. The system is implemented as a server-rendered monolith: a single Spring Boot application serves both the REST API and the Thymeleaf-based frontend, with all persistent state held in a PostgreSQL database.

The backend follows a layered architecture with clear separation of concerns: Controller layer (HTTP routing and view resolution), Service layer (business logic), Repository layer (data access via Spring Data JPA), and Model layer (JPA entities). Spring Security manages authentication via OAuth 2.0 (Google and GitHub) and enforces role-based access control across STANDARD, VERIFIED, and ADMIN roles. The frontend is rendered server-side using Thymeleaf templates, supplemented with Vanilla JavaScript for dynamic interactions. Containerisation is handled by Docker Compose, which orchestrates the Spring Boot application and the PostgreSQL database as isolated services.

4.1.2. Technology Stack

Backend:

Component	Technology / Version
Language	Java 17
Framework	Spring Boot 3.x
Web layer	Spring MVC (server-rendered, Thymeleaf views)
Data access	Spring Data JPA / Hibernate ORM
Security & Auth	Spring Security 6 + OAuth2 Client (Google, GitHub)
Database driver	PostgreSQL JDBC (pg)
Email	Spring Boot Starter Mail (Jakarta Mail / SMTP)
Input validation	Jakarta Bean Validation (Hibernate Validator)
Actuator	Spring Boot Actuator (health probe for Docker healthcheck)
Language	Java 17

Frontend:

Component	Technology / Version
Templating engine	Thymeleaf 3 (server-side rendering)
Markup	HTML5
Styles	CSS3 (custom, no external framework)
Scripting	Vanilla JavaScript (ES6+)

Build & Deployment:

Component	Technology / Version
Build tool	Apache Maven 3.x (pom.xml)
Containerisation	Docker + Docker Compose
Base images	maven:3-eclipse-temurin-17 (build stage), eclipse-temurin:17-jre-alpine (runtime)
Database	PostgreSQL 15 (postgres:15-alpine)
Production domain	listitup.duckdns.org (DuckDNS dynamic DNS)
SSL/TLS	Let's Encrypt certificate (via reverse proxy / host-level configuration)

4.1.3. Artifacts List**A. Source Code Artifacts****Entity Layer** (backend/src/main/java/com/listitup/api/model/)

File	Type	Description	Dependencies
User.java	JPA Entity	Platform user. Holds id, username, email, biography, profilePicture, role enum (STANDARD / VERIFIED / ADMIN), OAuth2 provider id, createdAt.	—
CuratedList.java	JPA Entity	A named, ordered list owned by a User. Fields: id, title, description, coverPhoto, visibility (PUBLIC / PRIVATE), createdAt. ManyToOne → User; OneToMany → Item, Comment, Like.	User, Category
Item.java	JPA Entity	A single entry inside a CuratedList. Fields: id, title, description, externalUrl, photo, positionIndex. ManyToOne → CuratedList.	CuratedList
Category.java	JPA Entity	Taxonomy for lists. Fields: id, name, icon. OneToMany → CuratedList.	—
Comment.java	JPA Entity	User comment on a list. Fields: id, text, timestamp. ManyToOne → User, CuratedList.	User, CuratedList
Like.java	JPA Entity	Like given by a User to a CuratedList. Fields: id, timestamp. Unique constraint on (user, list).	User, CuratedList
Follow.java	JPA Entity	Directed follow edge. Fields: id, follower (User), following (User), timestamp. Represents User→User M:N.	User
Notification.java	JPA Entity	In-app notification. Fields: id, message, isRead, timestamp, type. ManyToOne → User (recipient).	User
Report.java	JPA Entity	Content report. Fields: id, reason, status enum (OPEN / REVIEWED / RESOLVED), timestamp. ManyToOne → User (reporter).	User

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Repository Layer (backend/src/main/java/com/listitup/api/repository/)

File	Type	Description	Dependencies
UserRepository.java	JPA Repository	CRUD + custom queries: findByEmail, findByUsername, search by partial name.	User
CuratedListRepository.java	JPA Repository	CRUD + paginated queries for feed (public lists by recency/relevance) and search by keyword and category.	CuratedList, Category
ItemRepository.java	JPA Repository	CRUD + findByCuratedListOrderByPositionIndex for ordered item retrieval.	Item, CuratedList
CategoryRepository.java	JPA Repository	CRUD + findAll for category seed data.	Category
CommentRepository.java	JPA Repository	CRUD + findByCuratedListOrderByTimestampDesc.	Comment, CuratedList
LikeRepository.java	JPA Repository	CRUD + existsByUserAndCuratedList + countByCuratedList.	Like, User, CuratedList
FollowRepository.java	JPA Repository	CRUD + findByFollower, findByFollowing for social graph traversal.	Follow, User
NotificationRepository.java	JPA Repository	CRUD + findByRecipientOrderByTimestampDesc + markAllAsRead.	Notification, User
ReportRepository.java	JPA Repository	CRUD + findByStatus for admin moderation queue.	Report

Service Layer (backend/src/main/java/com/listitup/api/service/)

File	Type	Description	Dependencies
UserService.java	Service	Profile management: update bio/picture, role promotion, GDPR export and account deletion.	UserRepository
ListService.java	Service	List CRUD, visibility toggling, cover photo handling (upload to /app/uploads volume), feed and search pagination.	CuratedListRepository, CategoryRepository, ItemRepository
ItemService.java	Service	Item CRUD and reordering (swap positionIndex). Triggers link-metadata enrichment via OgService.	ItemRepository, CuratedListRepository, OgService
OgService.java	Service	Fetches Open Graph metadata (title, description, image) for item external URLs using jsoup.	—
SocialService.java	Service	Follow/unfollow, like/unlike, save-to-collection, comment CRUD. Triggers notifications after each social event.	FollowRepository, LikeRepository, CommentRepository, NotificationRepository
SearchService.java	Service	Delegates keyword + category search queries to CuratedListRepository with Pageable; returns Page<CuratedList>.	CuratedListRepository
NotificationService.java	Service	Creates in-app Notification records; enqueues email dispatch via EmailService.	NotificationRepository, EmailService
EmailService.java	Service	Sends transactional emails (like, comment, new-post) via JavaMailSender (SMTP). Template-rendered using Thymeleaf.	—
ReportService.java	Service	Accepts user reports; exposes admin query for open-report queue; updates report status.	ReportRepository
AnalyticsService.java	Service	Scheduled task (@Scheduled, daily at 03:00 UTC) aggregates viewCount, likeCount, saveCount, linkClickCount per list into analytics_snapshot table. Dashboard reads from snapshot.	CuratedListRepository, LikeRepository
AdminService.java	Service	Admin operations: content deletion (list, item, comment), report resolution.	CuratedListRepository, ItemRepository, CommentRepository, ReportRepository

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Security Layer (backend/src/main/java/com/listitup/api/security/)

File	Type	Description	Dependencies
SecurityConfig.java	Security Config	Defines Spring Security filter chain: OAuth2 login endpoints, CSRF configuration, role-based access rules (/admin/** → ADMIN; /analytics → VERIFIED/ADMIN; all others authenticated or public).	CustomOAuth2UserService
CustomOAuth2UserService.java	OAuth2 Service	Extends DefaultOAuth2UserService. On successful OAuth2 callback: looks up or creates User by email, maps provider attributes (Google/GitHub) to the User entity, sets default role.	UserRepository
OAuth2AuthenticationSuccessHandler.java	Auth Handler	Post-login redirect: sends authenticated user to /feed. Stores current user in session.	—

Controller Layer (backend/src/main/java/com/listitup/api/controller/)

File	Type	Description	Dependencies
HomeController.java	MVC Controller	GET / → landing page or redirect to /feed if authenticated.	—
FeedController.java	MVC Controller	GET /feed → paginated public-list feed for authenticated users, personalised by followed creators.	ListService, SocialService
ListController.java	MVC Controller	CRUD for CuratedList: GET/POST /lists/new, GET/POST /lists/{id}/edit, POST /lists/{id}/delete. Handles visibility toggle.	ListService
ListDetailController.java	MVC Controller	GET /lists/{id} → list detail view with items, comments, like/save status.	ListService, SocialService
ItemController.java	MVC Controller	POST /lists/{id}/items (add), POST /lists/{id}/items/{itemId}/edit, POST /lists/{id}/items/{itemId}/delete, POST /lists/{id}/items/reorder.	ItemService
SearchController.java	MVC Controller	GET /search?q=&category=&sort= → paginated search results.	SearchService
ProfileController.java	MVC Controller	GET /profile/{username}, POST /profile/edit, GET/POST /profile/follow/{userId}.	UserService, SocialService
SocialController.java	MVC Controller	POST /lists/{id}/like, POST /lists/{id}/save, POST /lists/{id}/comments, DELETE /comments/{id}.	SocialService
NotificationController.java	MVC Controller	GET /notifications, POST /notifications/mark-read.	NotificationService
AnalyticsController.java	MVC Controller	GET /analytics → creator dashboard (VERIFIED/ADMIN only). Supports ?sort=&order= params.	AnalyticsService
AdminController.java	MVC Controller	GET /admin → admin panel (ADMIN only). POST /admin/lists/{id}/delete, /admin/comments/{id}/delete, /admin/reports/{id}/resolve.	AdminService, ReportService
GdprController.java	MVC Controller	GET /profile/export → GDPR data export as JSON. POST /profile/delete → account deletion.	UserService

Web Forms (backend/src/main/java/com/listitup/api/dto/)

File	Type	Description	Dependencies
CreateListForm.java	DTO / Form	Binds HTML form for list creation/editing: title, description, categoryId, visibility.	CuratedList
CreateItemForm.java	DTO / Form	Binds HTML form for item creation/editing: title, description, externalUrl, photo (multipart).	Item
EditProfileForm.java	DTO / Form	Binds profile edit form: username, biography, profilePicture (multipart).	User
ReportForm.java	DTO / Form	Binds report submission form: targetType (list/item/comment), targetId, reason.	Report
SearchForm.java	DTO / Form	Binds search query parameters: q, category, sort, page.	—

Application Entry

File	Type	Description	Dependencies
ListItUpApplication.java	Spring Boot App	@SpringBootApplication entry point. Bootstraps the Spring context, enables JPA repositories, scheduling (@EnableScheduling for AnalyticsService), and multipart file upload support.	All beans

B. Configuration Files

File	Type	Description	Used By
backend/pom.xml	Maven POM	Declares all Maven dependencies (Spring Boot parent, Spring Security, Spring Data JPA, Thymeleaf, PostgreSQL driver, Actuator, Mail, Bean Validation, jsoup) and build plugins (maven-surefire-plugin, spring-boot-maven-plugin for fat-JAR packaging).	Maven build
backend/Dockerfile	Dockerfile	Two-stage build: stage 1 uses maven:3-eclipse-temurin-17 to compile and package the fat JAR (mvn package -DskipTests); stage 2 uses eclipse-temurin:17-jre-alpine as minimal runtime, copies the JAR, exposes port 8080.	Docker
docker-compose.yml	Docker Compose	Defines two services: db (postgres:15-alpine, volume pgdata, healthcheck pg_isready) and backend (built from ./backend/Dockerfile, depends_on db healthy, port 8082:8080, env vars injected from .env, volume uploads data:/app/uploads, healthcheck via Actuator /actuator/health). Resource limits: db 0.5 CPU / 256 MB, backend 1.0 CPU / 512 MB.	Docker Compose
backend/src/main/resources/application.properties	Spring Config	Configures datasource URL, JPA dialect (PostgreSQL), Hibernate DDL auto (validate in prod), Spring Security OAuth2 client registrations (Google, GitHub), Thymeleaf caching, multipart max file size, Actuator endpoints exposure, mail SMTP settings, and upload directory path.	Spring Boot
backend/src/main/resources/application-prod.properties	Spring Config (prod)	Production-profile overrides: disables Thymeleaf dev-mode caching, sets logging level to WARN, enforces HTTPS session cookie. Activated via SPRING_PROFILES_ACTIVE=prod env var in docker-compose.yml.	Spring Boot
.env.example	Env template	Documents required environment variables with comments: GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET (from Google Cloud Console), GITHUB_CLIENT_ID, GITHUB_CLIENT_SECRET (from GitHub OAuth Apps), SMTP_PASSWORD. Committed to git; actual .env is git-ignored.	Docker Compose / ops
.gitignore	Git config	Excludes .env, target/, *.class, IDE project files, and uploads/ directory from version control.	Git

C. Resource Files

Database Initialization

File	Type	Description
backend/src/main/resources/db/schema.sql	SQL DDL	Creates all tables (users, curated_lists, items, categories, comments, likes, follows, notifications, reports, analytics_snapshots) with constraints and indices. Run on first start (spring.sql.init.mode=always in dev, never in prod where Hibernate handles schema).
backend/src/main/resources/db/data.sql	SQL seed	Seeds the categories table with the initial set of categories (e.g., Movies, Books, Games, Music, Food, Travel, Tech, Sports) used in the feed and search filter.

Public Templates (backend/src/main/resources/templates/)

File	Type	Description
index.html	Thymeleaf	Landing page (unauthenticated). Hero section, value proposition, OAuth login buttons (Google, GitHub).
feed.html	Thymeleaf	Authenticated home feed. Paginated card grid of public lists; personalised results from followed creators shown first.
list-detail.html	Thymeleaf	Full list view. Displays ordered items, like/save controls, comment thread, report button. Edit/delete shown to list owner.
list-form.html	Thymeleaf	Create/edit list form. Fields: title, description, cover photo upload, category select, visibility toggle. Shared by create and edit flows.
item-form.html	Thymeleaf	Add/edit item form. Fields: title, description, external URL (with live OG-preview fetch), optional photo upload. Embedded in list-detail via Thymeleaf fragment.
profile.html	Thymeleaf	Public profile page. Shows avatar, bio, verified badge (if applicable), public lists grid, follower/following counts, follow button.
profile-edit.html	Thymeleaf	Edit own profile form. Fields: username, biography, profile picture upload.
search.html	Thymeleaf	Search results page. Keyword input, category filter dropdown, sort dropdown (Relevance / Recency), paginated results grid.
notifications.html	Thymeleaf	Notification centre. Lists in-app notifications in reverse chronological order with mark-all-read button.
analytics.html	Thymeleaf	Creator analytics dashboard (VERIFIED/ADMIN only). Per-list table with sortable columns: Views, Likes, Saves, Link Clicks. Data sourced from analytics_snapshots.
error/403.html	Thymeleaf	Forbidden error page (Spring Security redirects here on access-denied).
error/404.html	Thymeleaf	Not-found error page.
fragments/navbar.html	Thymeleaf Fragment	Reusable navigation bar fragment: logo, search bar, notification bell, user avatar dropdown.
fragments/list-card.html	Thymeleaf Fragment	Reusable list card component included in feed.html and search.html.

Admin Templates

File	Type	Description
admin/dashboard.html	Thymeleaf	Admin panel (ADMIN role only). Tabs: Reports queue (open reports with resolve action), Content moderation (search & delete any list, comment, or item), User management (promote to VERIFIED, view roles).

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Static Resources (backend/src/main/resources/static/)

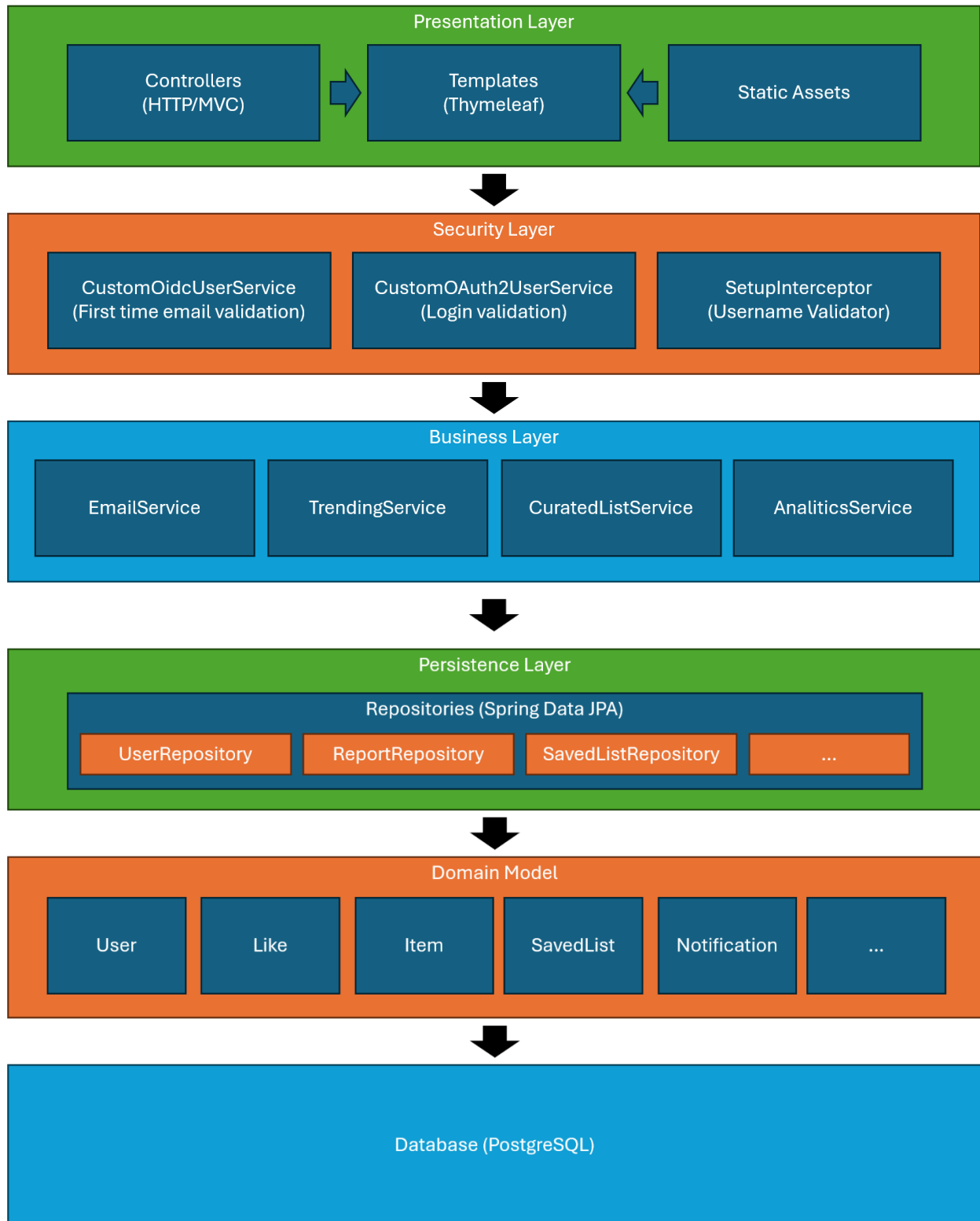
File / Directory	Type	Description
css/main.css	CSS	Global styles: CSS custom properties (colour tokens, spacing), typography, layout grid, responsive breakpoints, component styles (cards, buttons, forms, navbar, badges).
css/feed.css	CSS	Feed page-specific layout: masonry-style card grid.
css/list-detail.css	CSS	Detail page styles: item list, comment thread, sidebar.
css/admin.css	CSS	Admin panel table and tab styles.
js/og-preview.js	JavaScript	Fetches Open Graph metadata for item external URLs via GET /api/og?url= and populates preview fields inline in the item form.
js/item-reorder.js	JavaScript	Drag-and-drop reordering of items in the list-detail edit mode. POSTs new order to POST /lists/{id}/items/reorder.
js/notifications.js	JavaScript	Polls GET /notifications/count every 60 seconds to update the navbar notification bell badge without full page reload.
images/	Directory	Static image assets: logo, default avatar placeholder, category icons (SVG), OG fallback image.

D. Test Artifacts

Test artifacts are described in full in Section 4.2. The table below provides a summary index of the test source files.

File	Coverage Area	Test Count
UserServiceTest.java	User creation, role assignment, profile update, GDPR export, account deletion	12
ListServiceTest.java	List CRUD, visibility toggle, feed query, search pagination	15
ItemServiceTest.java	Item add/edit/delete, position reorder, OG fetch integration	10
SocialServiceTest.java	Follow/unfollow, like/unlike, save, comment CRUD, notification trigger	14
AnalyticsServiceTest.java	Daily aggregation scheduled task, snapshot correctness	6
ListControllerTest.java	MockMvc: GET/POST list endpoints, auth guards, redirect on unauthorised	12
AdminControllerTest.java	MockMvc: admin routes return 403 for STANDARD role, 200 for ADMIN	8
SecurityConfigTest.java	OAuth2 login flow, role-based access enforcement, CSRF	7
UserRepositoryTest.java	findByEmail, findByUsername, search query with @DataJpaTest	5
CuratedListRepositoryTest.java	Feed pagination, search by keyword + category with @DataJpaTest	7

4.1.4. Architecture & Relationships



The following structural relationships exist between the artefacts described above.

Inheritance

- No custom entity inheritance hierarchy is used. All JPA entities extend `Object` directly and are annotated with `@Entity`. Role differentiation (STANDARD, VERIFIED, ADMIN) is modelled as an enum field on `User` rather than a class hierarchy, keeping the schema flat and the codebase simple.

Composition

- CuratedList embeds a Category reference (ManyToOne, mandatory). Removing a Category would require reassigning its lists; no cascade delete is configured on this relationship.
- Follow uses two foreign keys (follower_id, following_id) pointing to the users table, implementing a self-referential M:N relationship without a separate join table entity in the ORM layer.

Associations

Association	Cardinality	Optional?	Cascade
User → CuratedList (creates)	1:N	Yes (user may have no lists)	CascadeType.ALL (delete user deletes lists)
CuratedList → Item	1:N	Yes (list may be empty)	CascadeType.ALL (delete list deletes items)
CuratedList → Comment	1:N	Yes	CascadeType.ALL
CuratedList → Like	1:N	Yes	CascadeType.ALL
User → Comment (writes)	N:1	No (comment must have author)	No cascade (delete user does not delete comments in MVP)
User → Like	N:1	No	No cascade
User → Follow (as follower)	N:M via Follow entity	Yes	CascadeType.ALL on Follow
User → Notification	1:N	Yes	CascadeType.ALL
User → Report (submits)	1:N	Yes	No cascade
CuratedList → Category	N:1	No (category is mandatory)	No cascade

4.1.5. Technical Requirements

Prerequisites:

- **Docker Engine** $\geq 24.x$ or **Docker Desktop** $\geq 4.x$ (includes Compose V2)
- **Docker Compose V2** (docker compose command, not docker-compose)
- **Git (to clone the repository)**
- **A .env file at the repository root populated from .env.example** (see Environment Variables below)
- Optional for local development without Docker: **JDK 17+**, **Apache Maven 3.9+**

Build & Run:

docker compose up -d --build

```
apueblar@raspberrypi:~/docker/ListItUp $ docker compose up -d --build
[+] Building 57.3s (17/17) FINISHED
=> [internal] load local bake definitions
=> == loading from stdin 230B
=> [internal] load build definition from Dockerfile
=> == transferring Dockerfile: 470B
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jre
=> [internal] load metadata for docker.io/library/maven:3.9.6-eclipse-temurin-17
=> [internal] load .dockerignore
=> == transferring context: 2B
=> [build 1/6] FROM docker.io/library/maven:3.9.6-eclipse-temurin:17-jre@sha256:39a1a50b1f90f9d6c2017f7b519b40eb47f6bd0020e097aac45a8c5c9f93004
=> [stage-1 1/3] FROM docker.io/library/eclipse-temurin:17-jre@sha256:d226a132511005726f330b0eca73004275000a03a0231e7b08512bc1500a7a
=> [internal] load build context
=> == transferring context: 0.46kB
=> CACHED [stage-1 2/3] WORKDIR /app
=> CACHED [build 2/6] WORKDIR /app
=> CACHED [build 3/6] COPY pom.xml .
=> CACHED [build 4/6] RUN mvn dependency:go-offline -B
=> CACHED [build 5/6] COPY src ./src
=> [build 6/6] RUN mvn package -DskipTests
=> [stage-1 3/3] COPY --from=build /app/target/*.jar app.jar
=> exporting to image
=> == exporting layers
=> == writing image sha256:da0220a014e0007f1794a0ef844920a9f2efad90b00a004324614200c00e79f5
=> == naming to docker.io/library/listitup-backend
=> == resolving provenance for metadata file
WARN[0057] Found orphan containers ([listitup_proxy]) for this project. If you removed or renamed this service in your compose file, you can r
[+] up 3/3
✔ Image listitup-backend      Built
✔ Container listitup_db       Healthy
✔ Container listitup_backend  Started
```

Run tests:

mvn test

Environment Variables:

.env — must be created from .env.example (never commit this file)

GOOGLE_CLIENT_ID=<your-google-client-id>

GOOGLE_CLIENT_SECRET=<your-google-client-secret>

GITHUB_CLIENT_ID=<your-github-client-id>

GITHUB_CLIENT_SECRET=<your-github-client-secret>

SMTP_PASSWORD=<app-password-for-smtp-account>

Ports:

Port	Service	Exposure	Notes
8080	Spring Boot (backend)	Internal only	Internal container port; not exposed to host in production.
8082	Spring Boot (backend)	Host-exposed	Mapped from container 8080. Used for local access and reverse-proxy forwarding.
5432	PostgreSQL (db)	Internal only	Accessible only within the listitup_net Docker bridge network.
443	HTTPS (reverse proxy)	Public	Production: SSL termination at host level; traffic forwarded to :8082.
80	HTTP (redirect)	Public	Production: redirects to HTTPS.

4.1.6. MVP Feature Coverage

The table below maps each functional requirement to its primary implementation artefacts and records the completion status as of the S4 submission.

ID	Feature	Primary Artefacts	Status
FR1	List creation and publishing	ListService, ListController, list-form.html	☑ Complete
FR2	Item management with drag-and-drop reorder	ItemService, ItemController, item-form.html, item-reorder.js	☑ Complete
FR3	Category-based discovery and keyword search	SearchService, SearchController, FeedController, search.html, feed.html	☑ Complete
FR4	Social interactions: follow, like, comment, save	SocialService, SocialController, ProfileController, list-detail.html, profile.html	☑ Complete
FR5	Content reporting and admin moderation panel	ReportService, AdminService, AdminController, admin/dashboard.html	☑ Complete
FR6	In-app and email notifications	NotificationService, EmailService, NotificationController, notifications.html, notifications.js	☑ Complete
FR7	Creator analytics dashboard (VERIFIED/ADMIN)	AnalyticsService, AnalyticsController, analytics.html	☑ Complete
—	OAuth2 authentication (Google + GitHub)	SecurityConfig, CustomOAuth2UserService, OAuth2AuthenticationSuccessHandler	☑ Complete
—	GDPR data export and account deletion	GdprController, UserService (/profile/export, /profile/delete)	☑ Complete
—	Open Graph link metadata enrichment	OgService, og-preview.js (live preview in item form)	☑ Complete

4.2. Tests

The documentation should cover the methods and test results of the specific system's element both from the functional and non-functional point of view. The scope and the description should conform to the specification of the requirements (F2). You should include a description of at least one test of each type of requirement (e.g. business logic, user interface, data exchange, etc.).

4.2.1. Requirements Tests

In this section, you should differentiate the description of tests from manual and automated tests. You should include scripts or test case scenarios in the report.

4.2.1.1. Functional requirements tests

FR1 – Create a List

List It Up Search [+ Create List](#) [Admin Panel](#) [M](#)

Create a New List [✕ Cancel](#)

List Title
Best spots in Budapest

Description
My favourite spots in the city of Budapest, Hungary

Category Can't find it? Propose a new one
Travel

Cover Photo (Optional)
Drag & Drop an image here or click to select

OR provide a URL: <https://hips.hearstapps.com/hmg-prod/images/budapest-danubio-parlamento-1552491234.jpg>

List Items [+ Add Item](#)

Item Title
Fisherman's Bastion

Description
Next to the Buda castle, the best place in Budapest for a nice view of the city. It's a beautiful historical monument

Photo (Optional)
Drag & Drop an image here or click to select

OR provide a URL: <https://www.ulysses.travel/wp-content/uploads/2020/07/Le-Bastion-des-Pe%CC%82cheur>

External URL (Optional)
<https://www.ulysses.travel/en/fishermans-bastion-budapest/>

[Save as Draft](#) [Publish List](#)

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List It Up

Search

+ Create List

Admin Panel

1

M



Travel

Best spots in Budapest

My favourite spots in the city of Budapest, Hungary

By miryam.merl · 2 views · 1 likes

Like

Save

Edit

Delete

1



Fisherman's Bastion

Next to the Buda castle, the best place in Budapest for a nice view of the city. It's a beautiful historical monument

[Visit Link →](#)

2



Ildikó Konyhája Hungarian Restaurant

Best place to eat after a long walk in the Buda side. After going to Buda castle, this is a must

Comments

Add a comment...

Post

No comments yet. Be the first to comment!

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FR2 – Manage Items (Add, edit, reorder, delete)

- Add

List It Up

 Search

+ Create List

Create a New List

× Cancel

List Title

My Favourite Andalusian Spots

Description

List of my favourite places in Andalusia, a must!

Category

Travel

Can't find it? Propose a new one

Cover Photo (Optional)

Drag & Drop an image here or click to select

OR provide a URL:

/uploads/be1810df-4f4d-4ac8-abf2-c72b6b74974d.webp

List Items

+ Add Item

☰ ▲ ▼

×

Item Title

e.g. The Matrix

Description

Why is this on your list?

Photo (Optional)

Drag & Drop an image here or click to select

OR provide a URL:

https://

External URL (Optional)

https://

Save as Draft

Publish List

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- Edit

List It Up

 Search

+ Create List





Edit List

✕ Cancel

List Title

My Favourite Andalusian Spots

Description

List of my favourite places in Andalusia, a must!

Category

Can't find it? Propose a new one

Travel

Cover Photo (Optional)

Drag & Drop an image here or click to select

OR provide a URL: /uploads/be1810df-4f4d-4ac8-abf2-c72b6b74974d.webp

List Items

+ Add Item

☰ ▲ ▼ ✕

Item Title

Mezquita-Cathedral of Córdoba

Description

The most beautiful colossal, UNESCO-listed monument in Andalusia, Spain.

Photo (Optional)

Drag & Drop an image here or click to select

OR provide a URL: /uploads/25ec226a-3942-4ef8-97e7-2c5234183570.jpg

External URL (Optional)

https://mezquita-catedraldecordoba.es/en/

☰ ▲ ▼ ✕

Item Title

Tinto River in Huelva

Description

The Río Tinto in Huelva, Spain, is a 100-kilometer-long river known for its surreal, rust-red waters and extreme acidity (pH 2).

Photo (Optional)

Drag & Drop an image here or click to select

OR provide a URL: /uploads/87c969cc-4bf8-4441-aeed-1f034f9d2dfc.jpg

External URL (Optional)

☰ ▲ ▼ ✕

Item Title

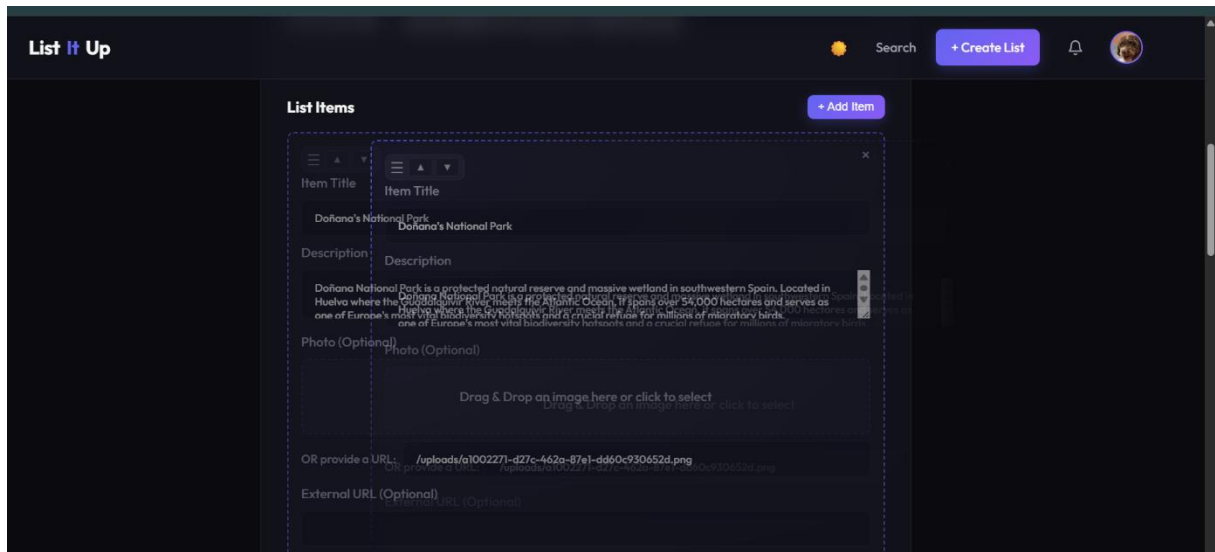
Puente Romano in Córdoba

Description

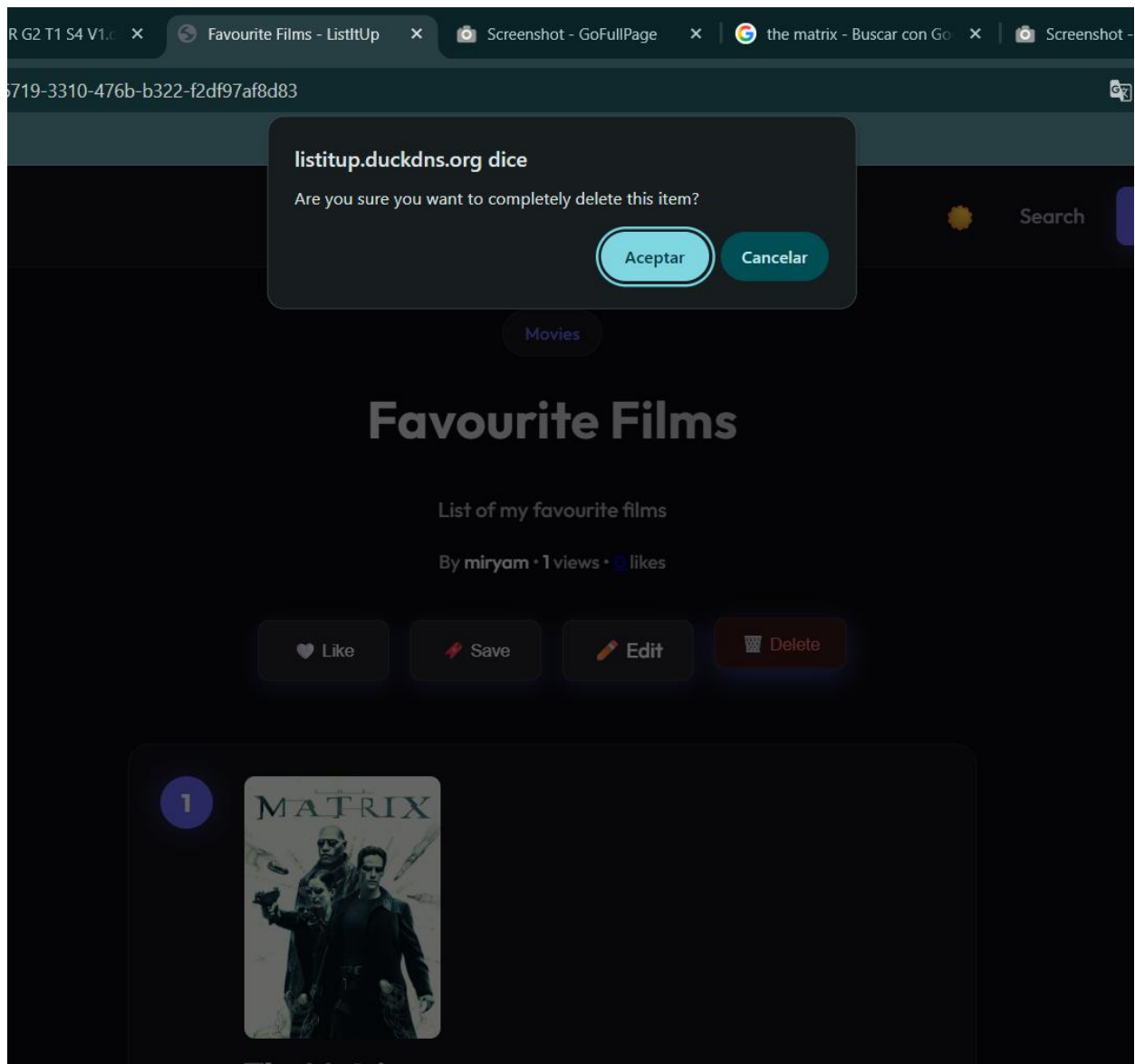
The Roman bridge of Córdoba (Puente Romano) is a historic 331-meter stone pedestrian bridge spanning the Guadalquivir River in southern Spain. Built originally in the 1st century BC, it served as the city's only river crossing for 2,000 years and is famous for its appearance as the Lona Bridge of Valantia in Game of Thrones.

Photo (Optional)

- Reorder

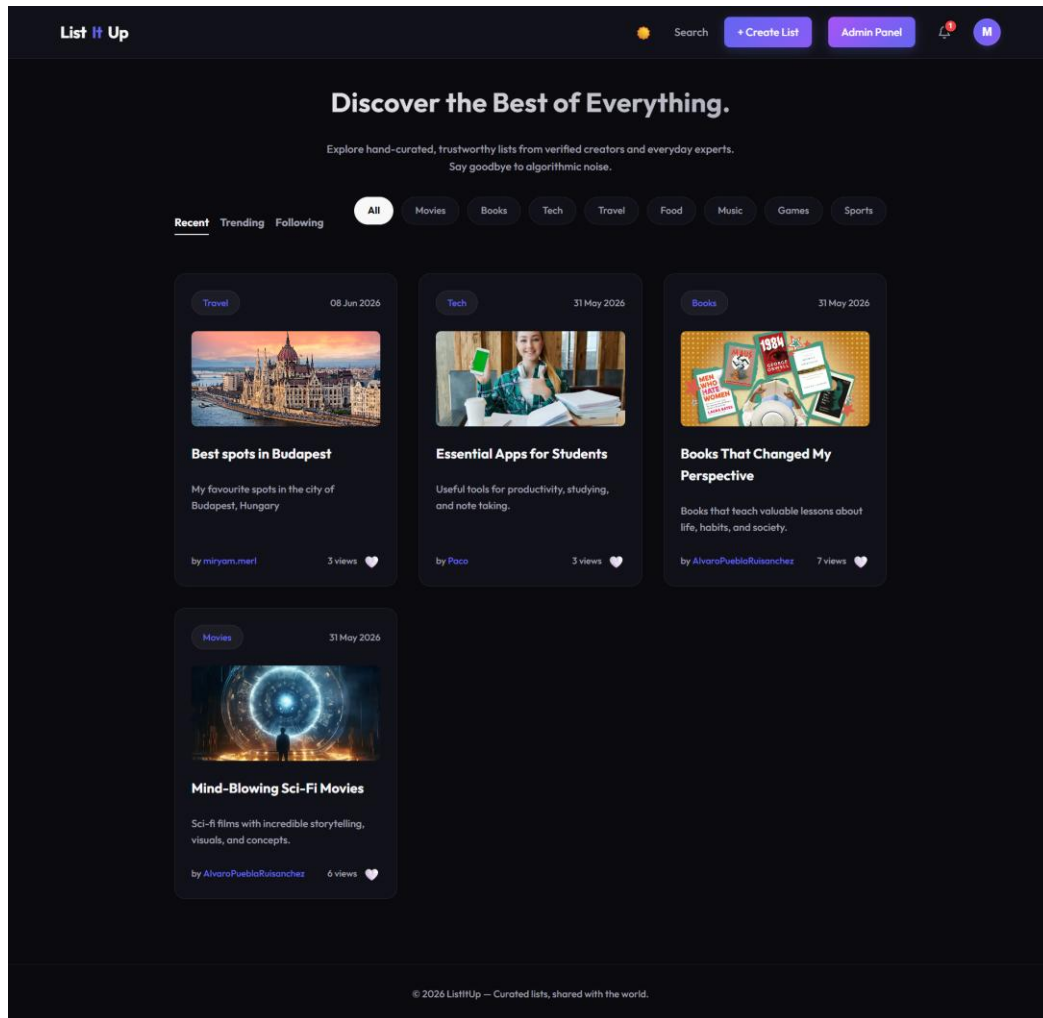


- Delete

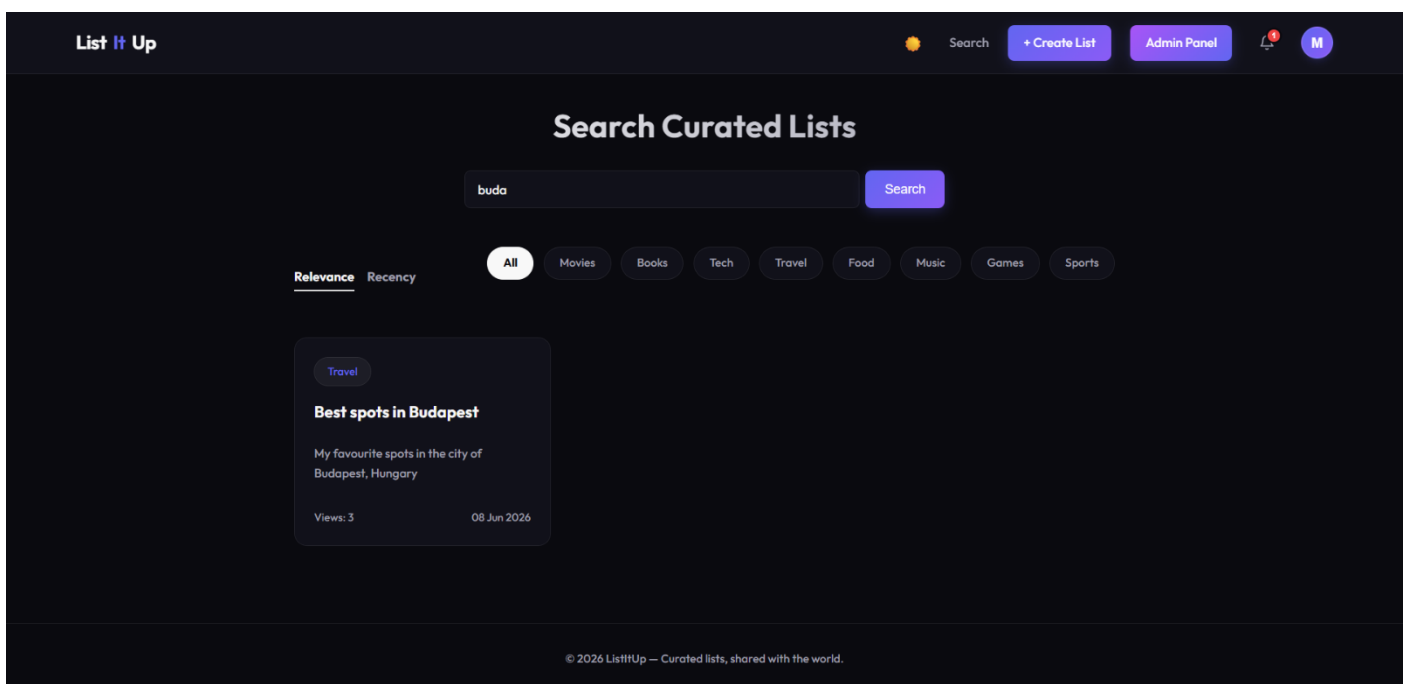


FR3 – Discovery and Search

- By Category

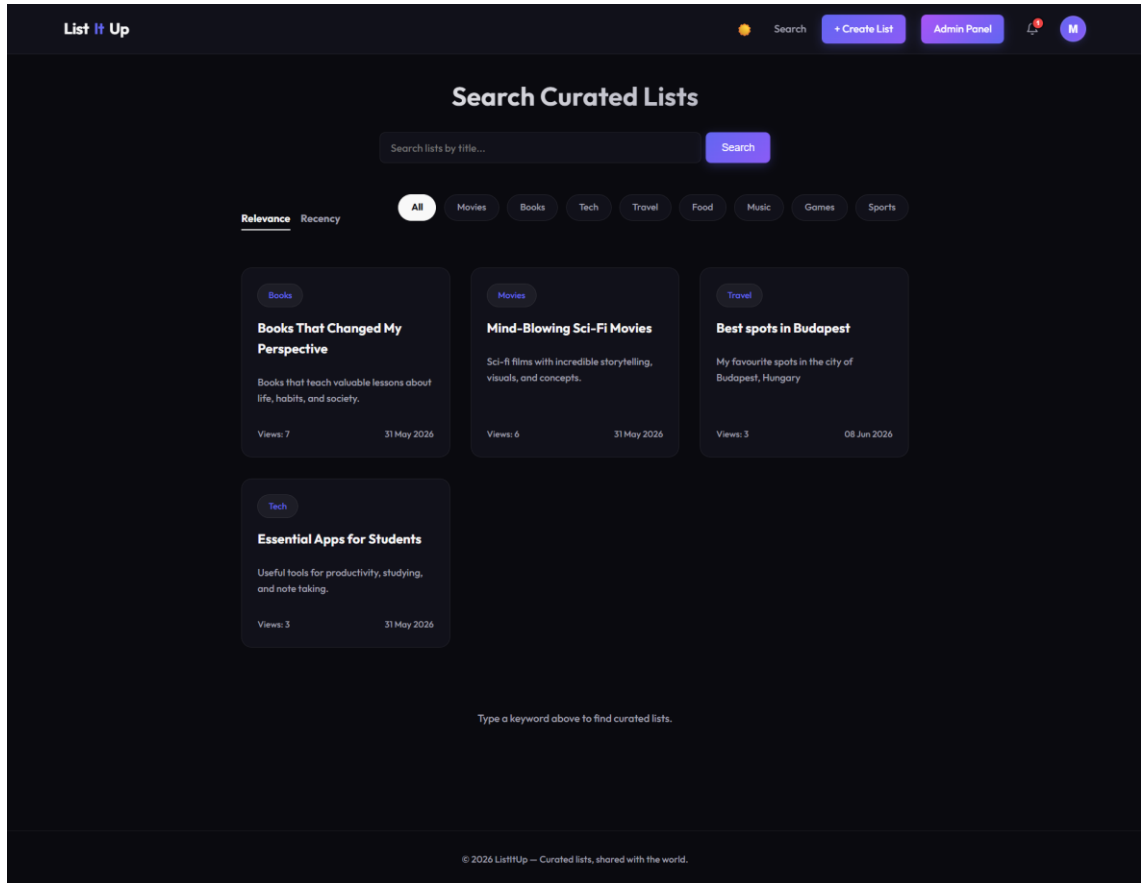


- By Keyword

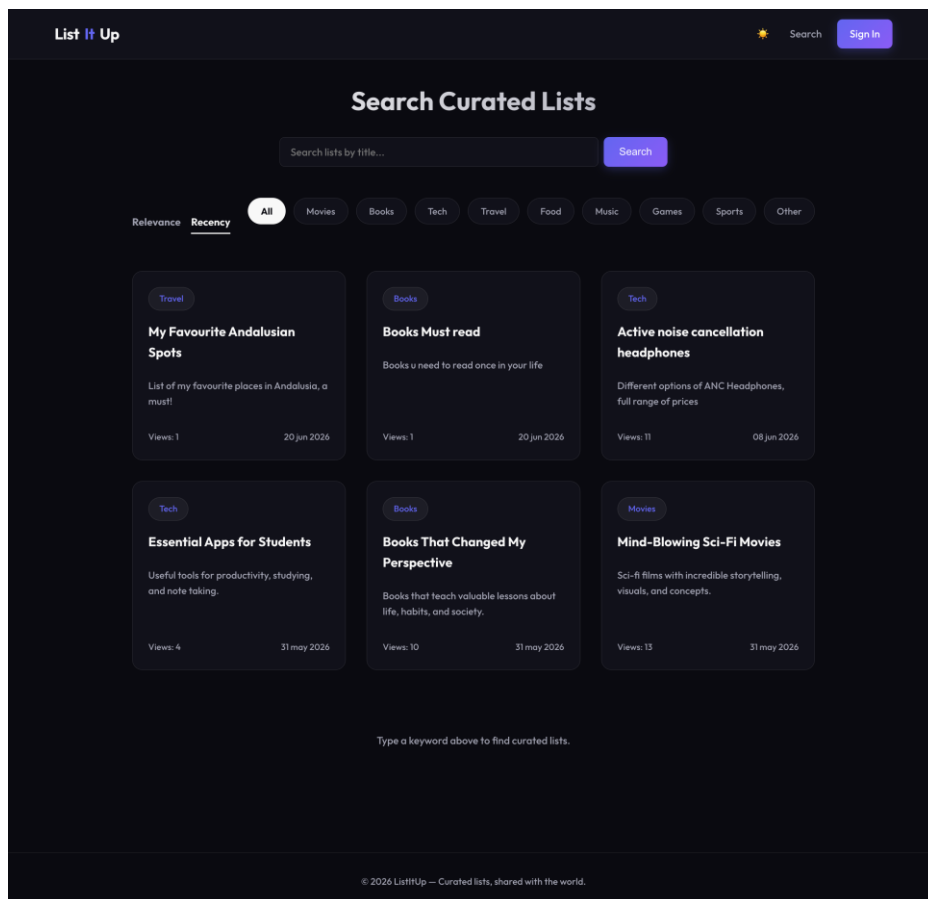


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- By Relevance

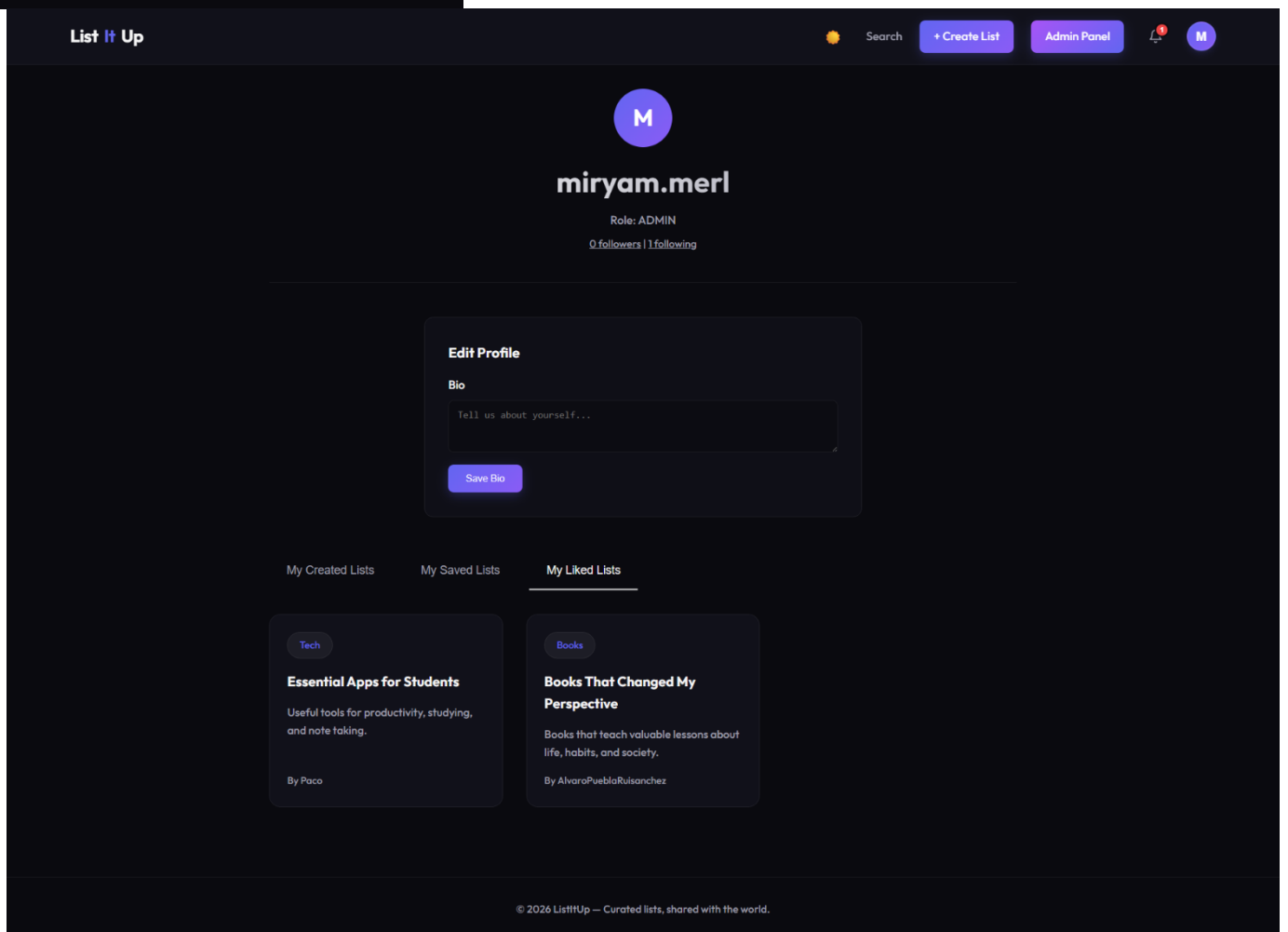
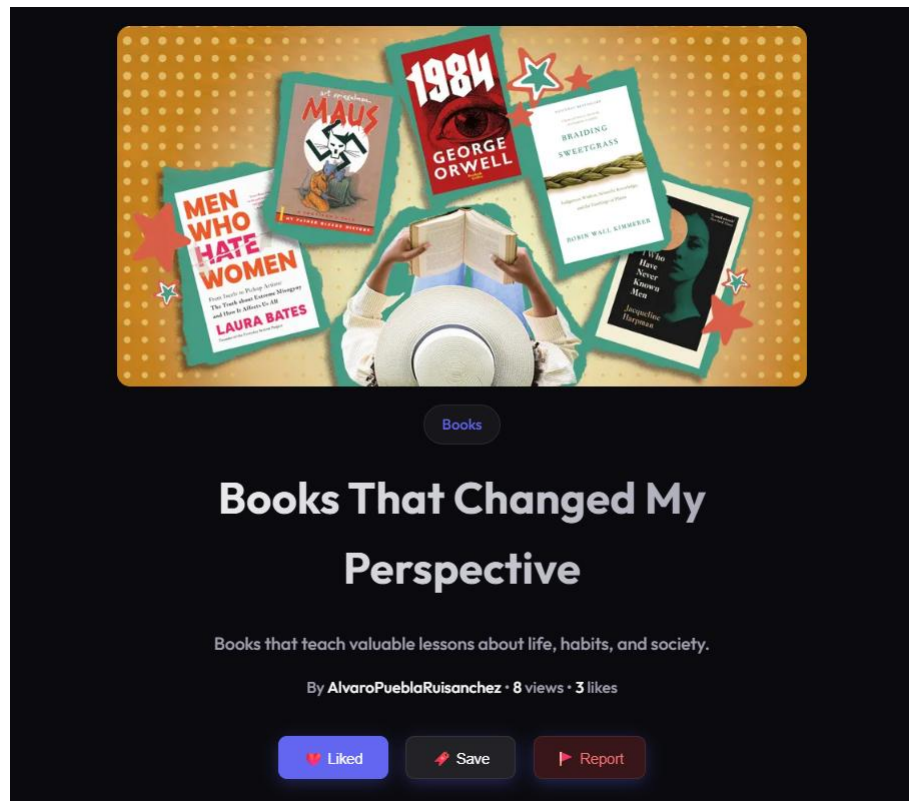
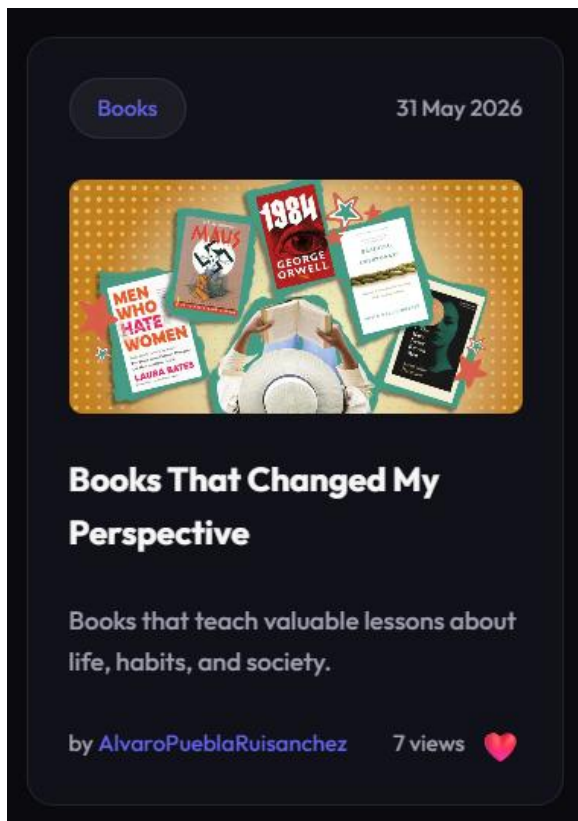


- By Recency



FR4 – Social Interactions

- Like Lists



- Folow

List It Up

Search

+ Create List

Admin Panel

1

M

A

AlvaroPueblaRuisanchez

VERIFIED

4 followers

0 following

Following

Joined May 2026

AlvaroPueblaRuisanchez's Lists

Books

Books That Changed My Perspective

Books that teach valuable lessons about life, habits, and society.

7 views

31 May 2026

Movies

Mind-Blowing Sci-Fi Movies

Sci-fi films with incredible storytelling, visuals, and concepts.

6 views

31 May 2026

© 2026 ListItUp — Curated lists, shared with the world.

- Post Comments

List It Up

Search

+ Create List

Admin Panel

1

M

MAUS

1984

MEN WHO HATE WOMEN

LAURA BAKI

Books

Books That Changed My Perspective

Books that teach valuable lessons about life, habits, and society.

By AlvaroPueblaRuisanchez · 8 views · 3 lists

Like

Save

Report

1

This Changes, Remarkable Beauty

Atomic Habits

James Clear

Atomic Habits

Practical strategies for building good habits and breaking bad ones.

Visit Link

Report Item

Comments

Add a comment...

Post

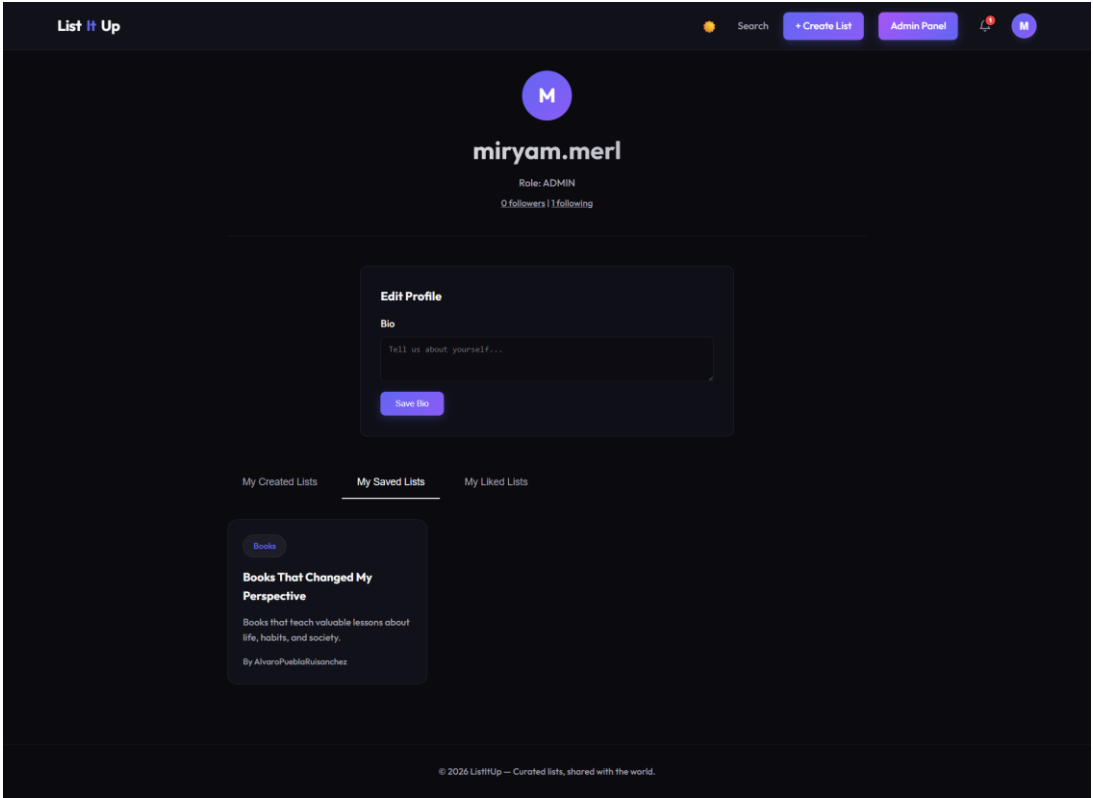
mlryam_mart

Amazing book! Atomic Habits teaches you what to do but it does this by teaching you the science behind it. Knowing the underlying science helps you shape your own habits effectively

08 Jun 2026, 08:59

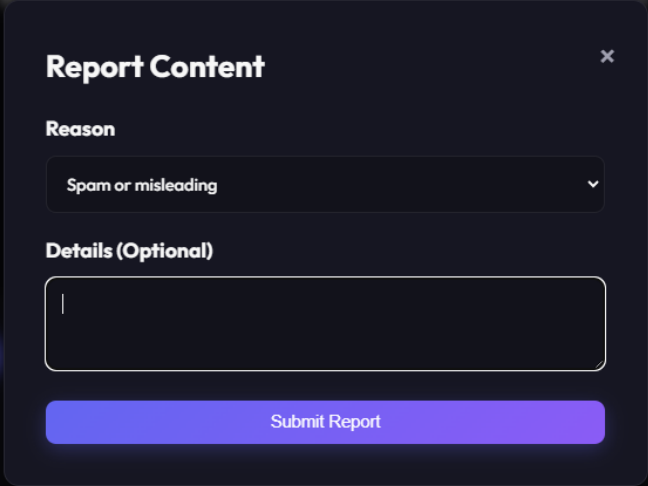
© 2026 ListItUp — Curated lists, shared with the world.

- Save Lists



FR5 – Reporting and Moderation

- Report List



A dark-themed modal window titled "Report Content" with a close button (X) in the top right corner. It contains a "Reason" dropdown menu currently set to "Spam or misleading", a "Details (Optional)" text input field, and a blue "Submit Report" button at the bottom.

- Report Item

1



Movie poster for Interstellar, featuring a man in a space suit standing on a rocky, icy planet surface. The title "INTERSTELLAR" is at the top, and a quote "A MUST-SEE MASTERPIECE" is at the bottom.

Interstellar

A journey through space and time to save humanity.

[Visit Link →](#)

[▶ Report Item](#)

- Report Comment

Comments

Add a comment...

Post

Mamawevo

ReportDelete

I hate you and your lists F*CK YOU!

08 Jun 2026 09:18

List It Up

Search

Create List

Admin Panel

M

Report Moderation

Review content reports filed by users, resolve active flag notifications, or dismiss content violations.

Users

Lists

Reports

Categories

Pending Reports

Date	Target Content	Reporter	Reason	Details	Actions
08 Jun 2026	Comment: "I hate you and your lists F*CK YOU!" On list: Best spots in Budapest	miryam.merl	HARASSMENT	Harassed the owner of the list	Delete Comment Resolve Dismiss
08 Jun 2026	Item: Interstellar On list: Mind-Blowing Sci-Fi Movies	miryam.merl	INAPPROPRIATE		Resolve Dismiss
08 Jun 2026	List: Mind-Blowing Sci-Fi Movies	miryam.merl	OTHER		Resolve Dismiss
08 Jun 2026	Item: Fisherman's Bastion On list: Best spots in Budapest	Mamawevo	INAPPROPRIATE	Very explicit	Resolve Dismiss

Resolved / Dismissed History

RESOLVED

List: Books That Changed My Perspective

01 Jun 2026

Reason: SPAMReporter: PueblaDetails: This list is badModerator: System

DISMISSED

List: Mind-Blowing Sci-Fi Movies

08 Jun 2026

Reason: SPAMReporter: miryam.merlDetails: NoneModerator: miryam.merl

DISMISSED

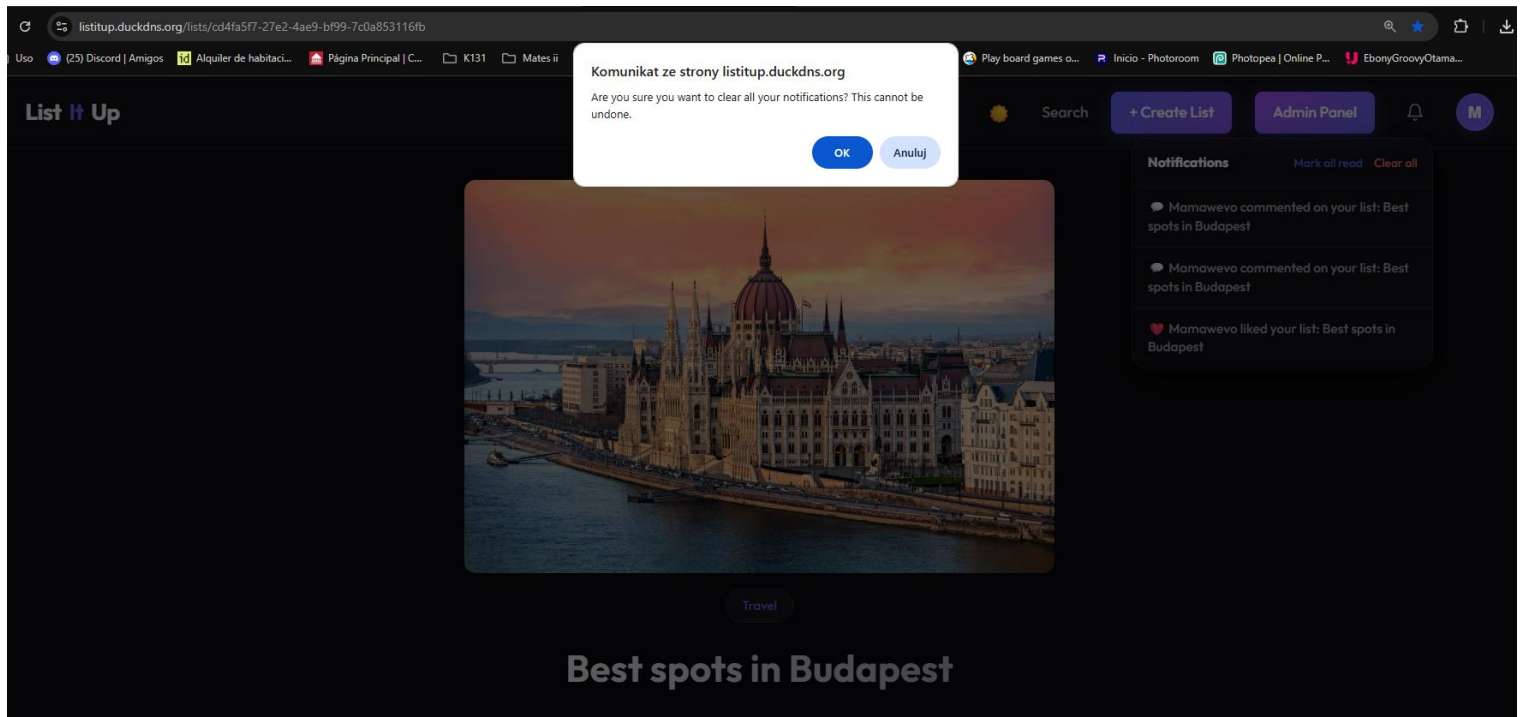
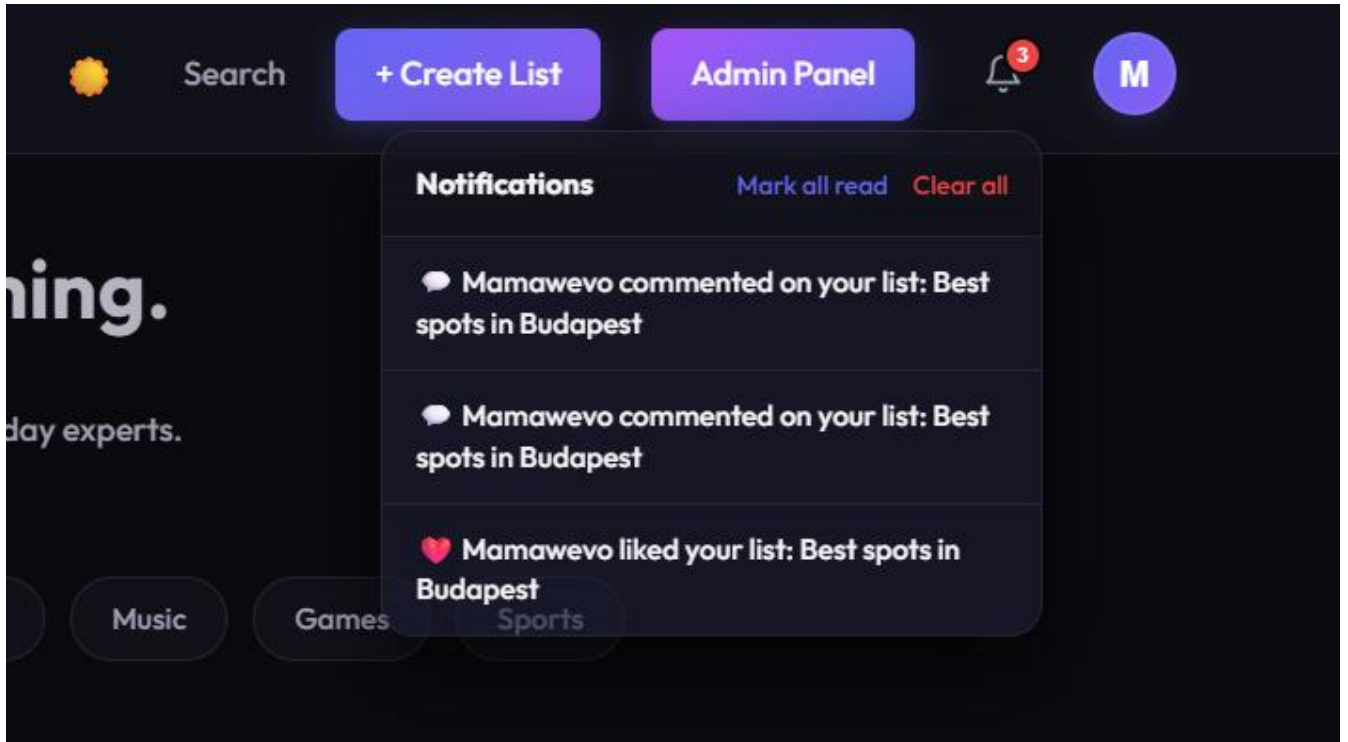
List: Books That Changed My Perspective

01 Jun 2026

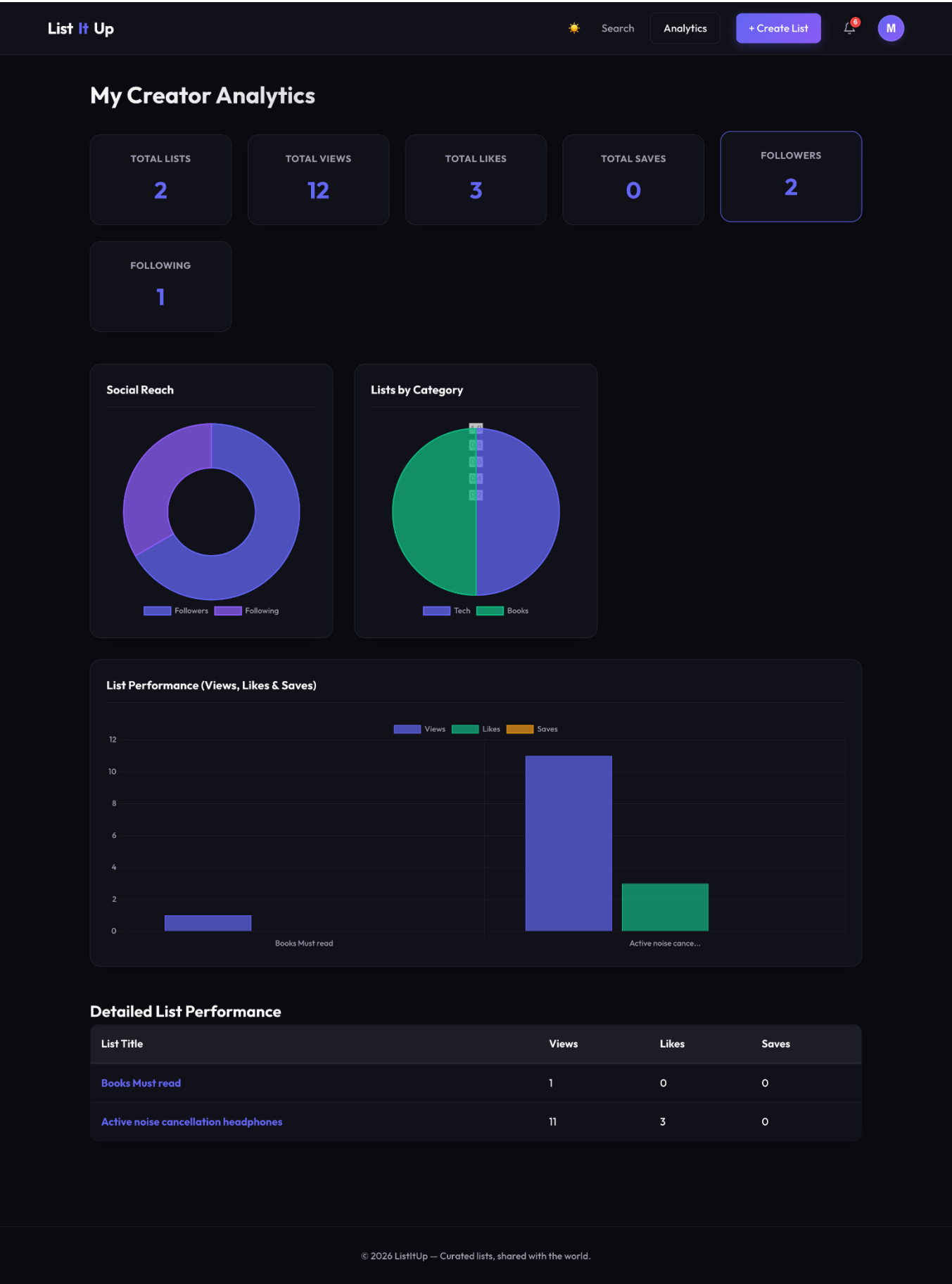
Reason: SPAMReporter: ApueblarDetails: Miss clickModerator: System

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FR6 – Notifications



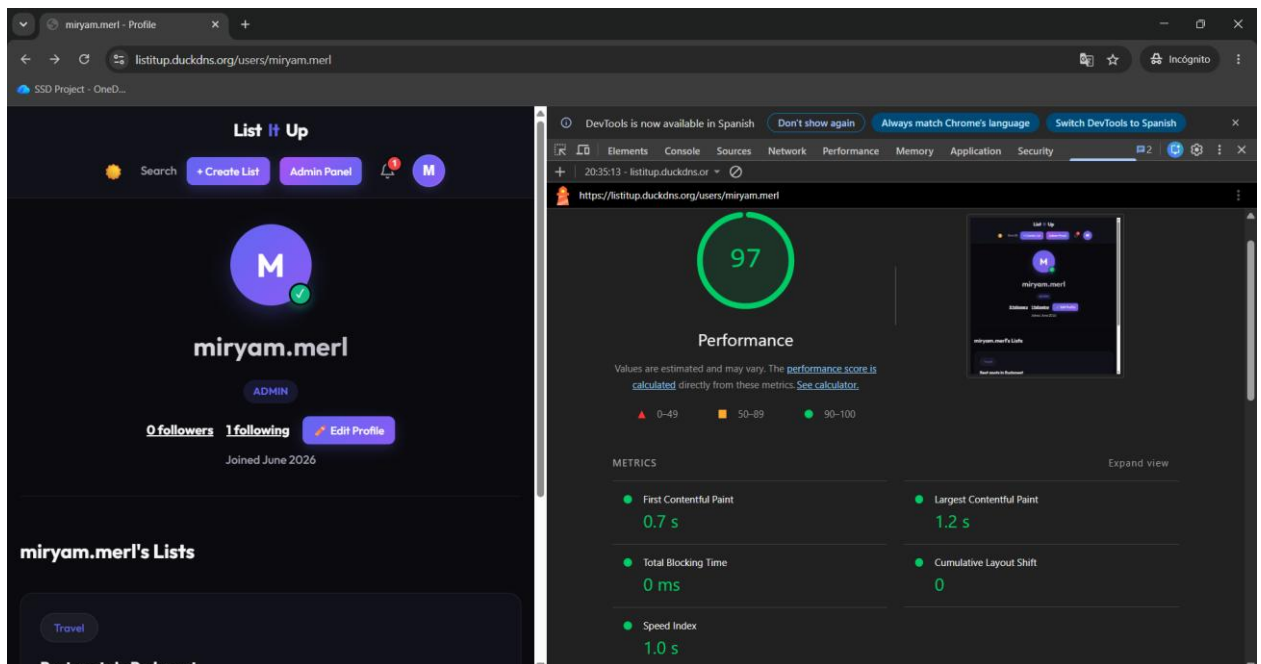
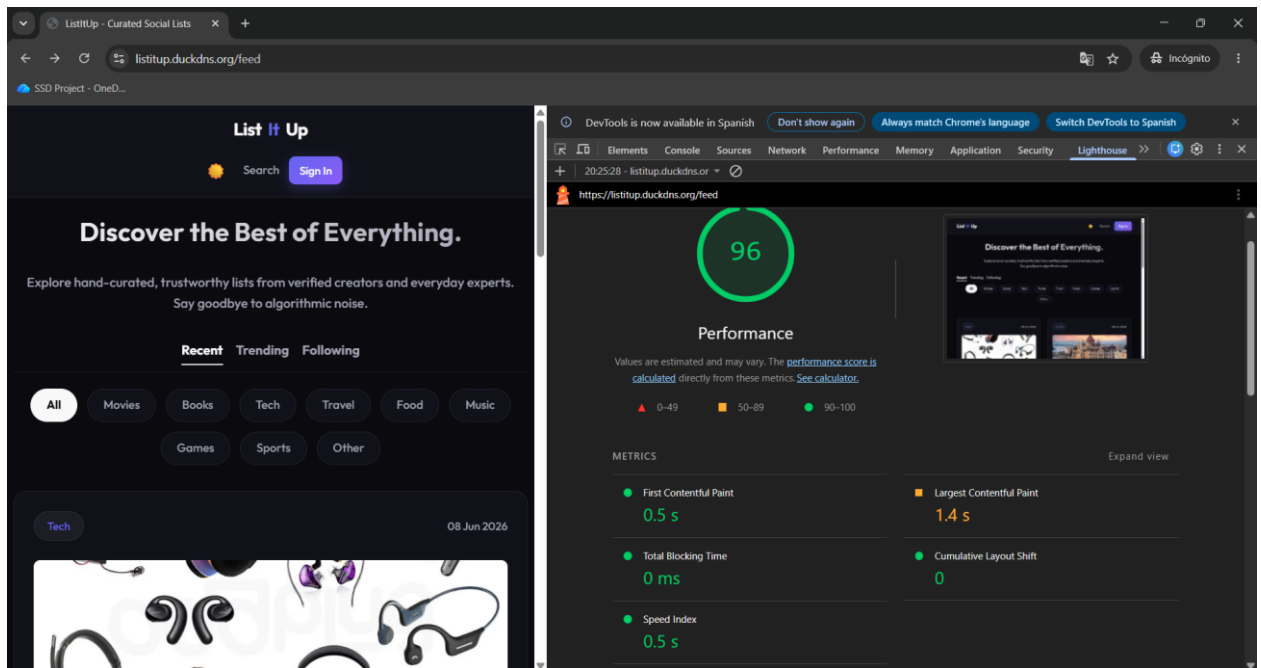
FR7 – Creator Analytics



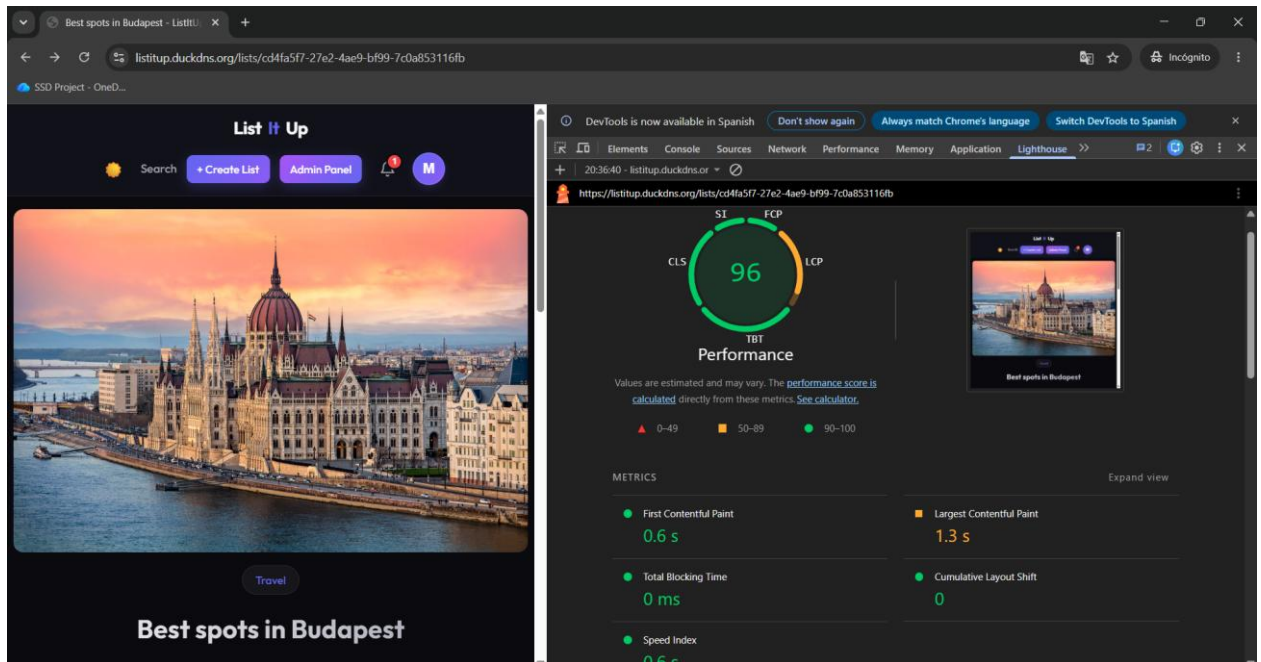
4.2.1.2. Non-Functional requirements tests

NF1 – Page Load time < 3 sec

- *Type:* Performance / Efficiency.
- *Description:* Primary pages must fully render within 3 seconds on a standard broadband connection, measured as Largest Contentful Paint (LCP) and Total Blocking Time (TBT).
- *Verification Method:* Google Chrome (Lighthouse).
- *Test Results:* The web successfully meets the requirement of fully rendering within 3 seconds. Across all three tested pages, the FCP consistently remained under 0.8 seconds, and the LCP never exceeded 1.4 seconds. Because the heaviest content on our core pages becomes fully visible in less than half the maximum allowed time, the application proves to be highly performant and responsive.

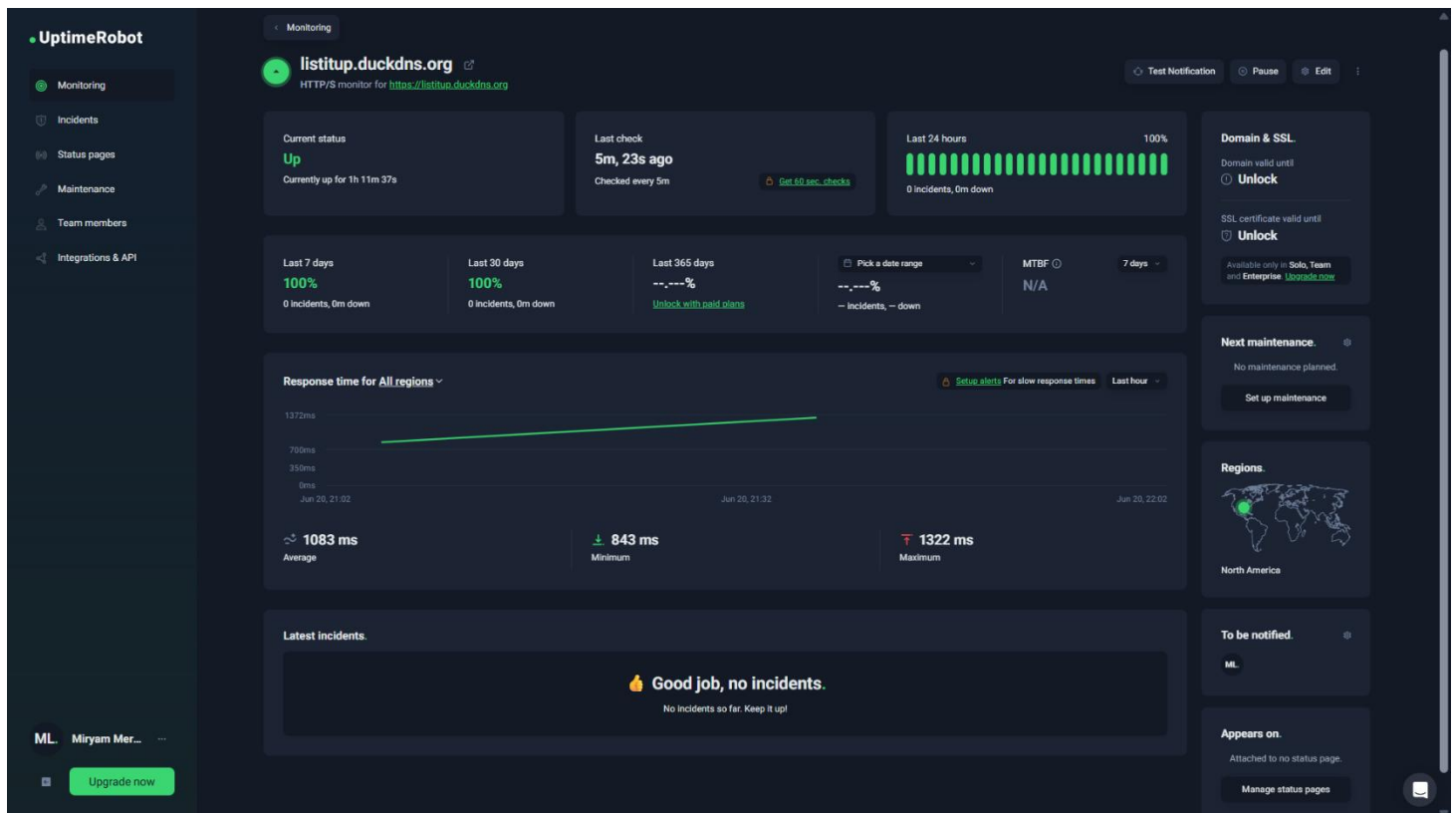


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NF2 – System Uptime $\geq 99.5\%$

- *Type:* Reliability / Availability.
- *Description:* The platform must maintain at least 99.5% uptime throughout the pilot period.
- *Verification Method:* UptimeRobot.
- *Test Results:* To verify the system's reliability, we configured an UptimeRobot to continuously track the production environment at `https://listitup.duckdns.org`. The monitoring results confirm that the platform successfully remained stable, maintaining an uptime of 100%, which exceeds the required 99.5% threshold established for the pilot period.



NF3 – HTTPS Encryption (>=TLS 1.2)

- **Type:** Security
- **Description:** All client-server communication must be encrypted using TLS 1.2 or higher. No sensitive data may be transmitted over plain HTTP.
- **Verification Method:** Browser Developer Tools and SSL Labs security test.
- **Test Results:** To verify that all client-server communication is securely encrypted, a two-part validation was performed. First, inspection via the browser's Developer Tools (Network tab) confirmed that all active requests are routed exclusively through HTTPS. Second, an external security audit was conducted using Qualys SSL Labs on the production domain (listitup.duckdns.org). The server achieved an overall rating of "A", confirming that it correctly enforces TLS 1.2 or higher and meets modern cryptographic standards.

HTTPs header:

The screenshot displays a web browser window with the URL `listitup.duckdns.org/search?q=&category=Movies&sort=`. The page content includes a search bar, category filters (All, Movies, Books, Tech, Travel, Food, Music, Games, Sports, Other), and a section titled "Mind-Blowing Sci-Fi Movies".

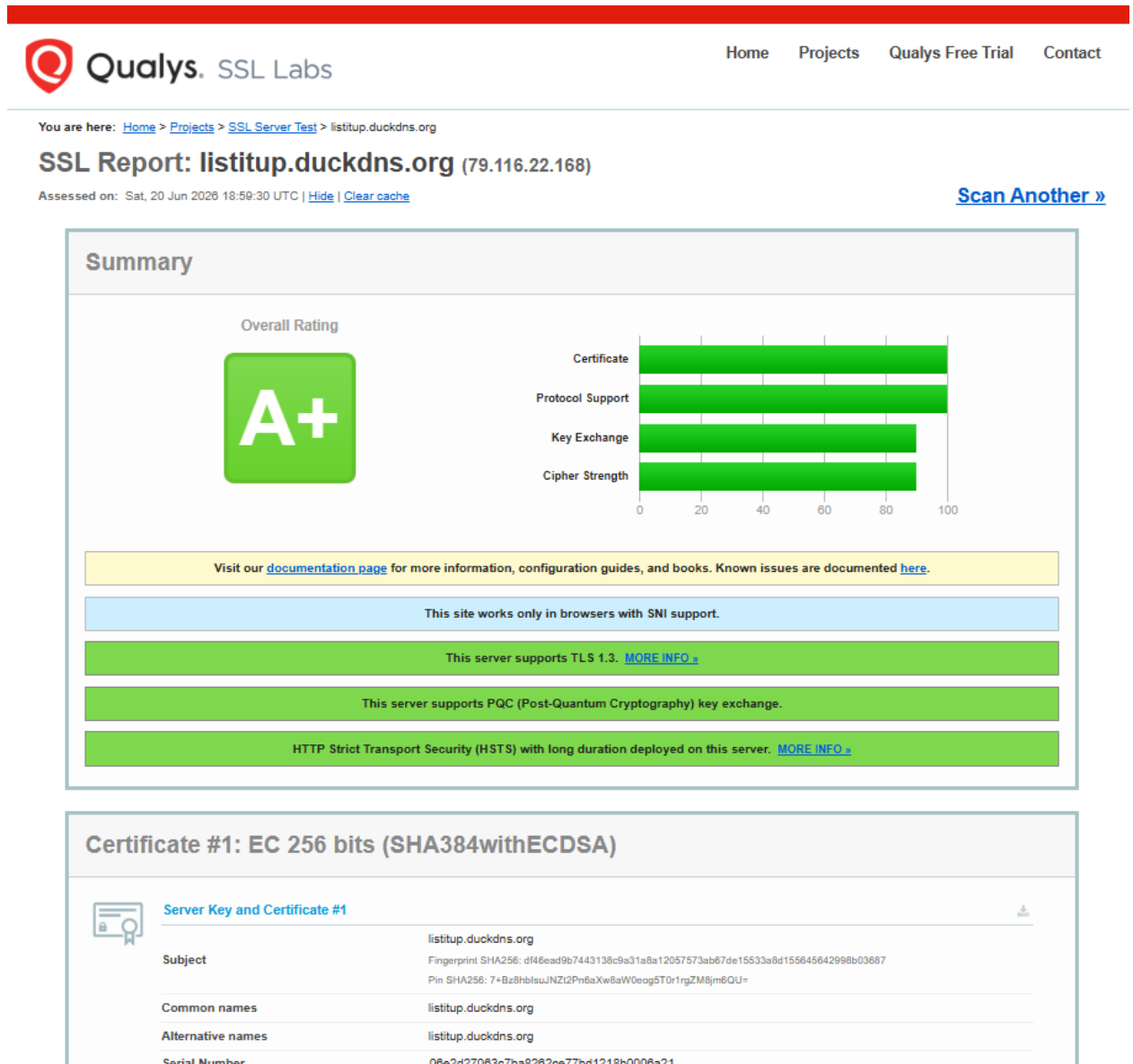
The browser's developer tools are open, showing the Network tab. A list of resources is displayed, including `search?q=&category=Movies...`, `css?family=interwght@400,5...`, `style.css?v=1.1`, `header.css?v=1.1`, `footer.css?v=1.1`, `theme-init.js`, `search.css`, and `app.js`. The selected resource is `search?q=&category=Movies...`, and its details are shown in the right pane.

The details pane shows the following information:

- General:**
 - Request URL: `https://listitup.duckdns.org/search?q=&category=Movies&sort=`
 - Request Method: `GET`
 - Status Code: `200 OK`
 - Remote Address: `79.116.22.168:443`
 - Referrer Policy: `no-referrer`
- Response headers:**
 - Alt-Svc: `h3="443"; ma=2592000`
 - Cache-Control: `no-cache, no-store, max-age=0, must-revalidate`
 - Content-Encoding: `gzip`
 - Content-Language: `en`
 - Content-Type: `text/html; charset=UTF-8`
 - Date: `Sat, 20 Jun 2026 18:56:15 GMT`
 - Expires: `0`
 - Pragma: `no-cache`
 - Referrer-Policy: `no-referrer`
 - Strict-Transport-Security: `max-age=31536000; includeSubDomains; preload`
 - Vary: `Origin`
 - Access-Control-Request-Method: `Vary`

The bottom status bar indicates 8 / 10 requests, 13.1 kB / 13.4 kB.

SSL Report:

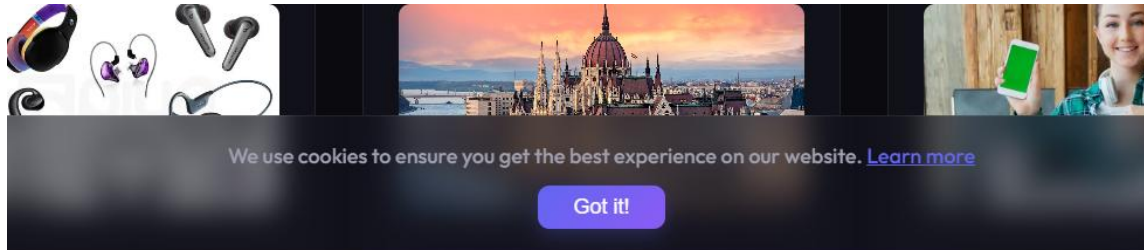
**NF4 – GDPR / Legal Compliance**

- *Type:* Legal / Privacy
- *Description:* The platform must comply with GDPR/RODO: users must be able to access, rectify, and erase their personal data. A cookie-consent banner must be displayed on first visit before any non-essential cookies are set.
- *Verification Method:* Cookie consent showing on incognito browsing session, manual testing of the profile management interface and account deletion workflow.
- *Test Results:* To verify compliance with GDPR/RODO regulations, a manual verification of user data privacy rights was conducted according to the specified criteria:
 - Cookie Consent: An incognito browsing session confirmed that a cookie-consent banner is prominently displayed upon the first visit, ensuring no non-essential cookies are set prior to user consent.

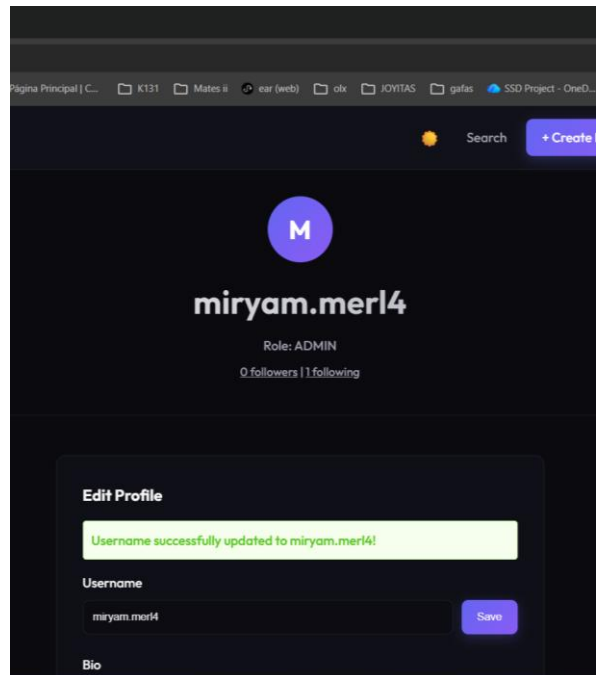
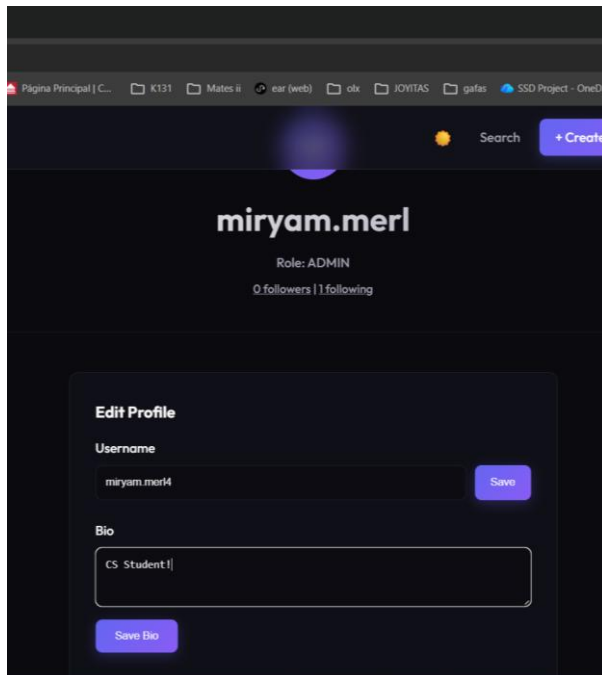
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- Data Access and Rectification: Manual testing of the profile management interface confirmed that users can successfully access and update their personal information at any time.
- Right to Erasure (Deletion): The account deletion workflow was manually executed. Subsequent login attempts asked for a new username.

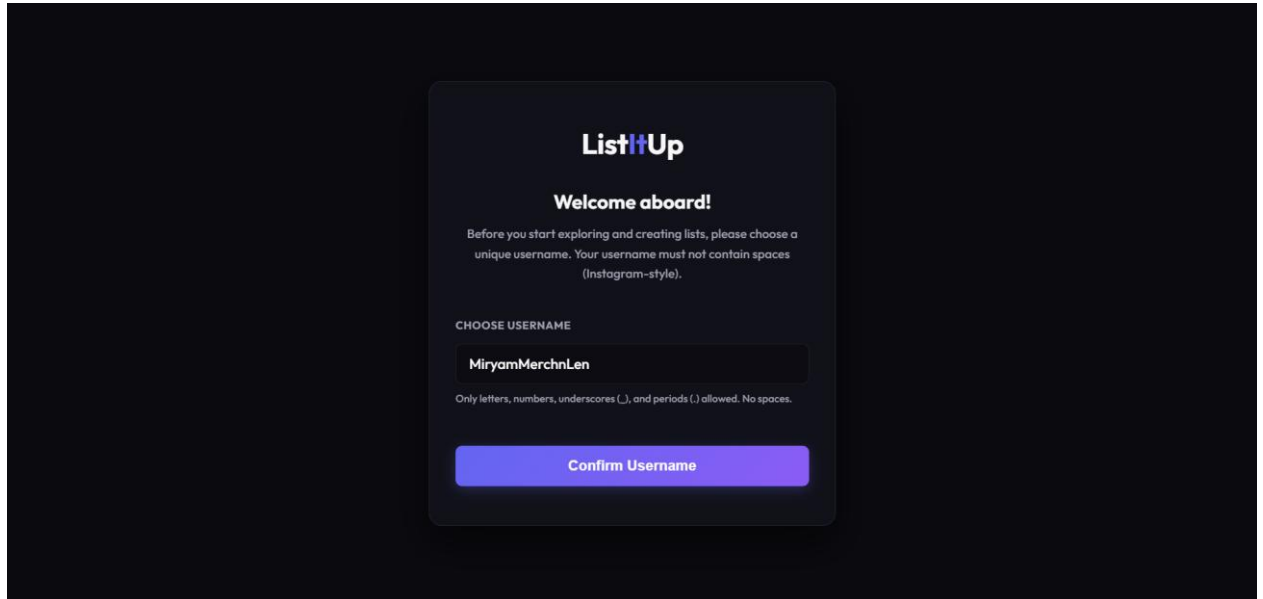
Cookie Consent:



Access and Rectify:



After Deletion:



NF5 - Task Completion Rate

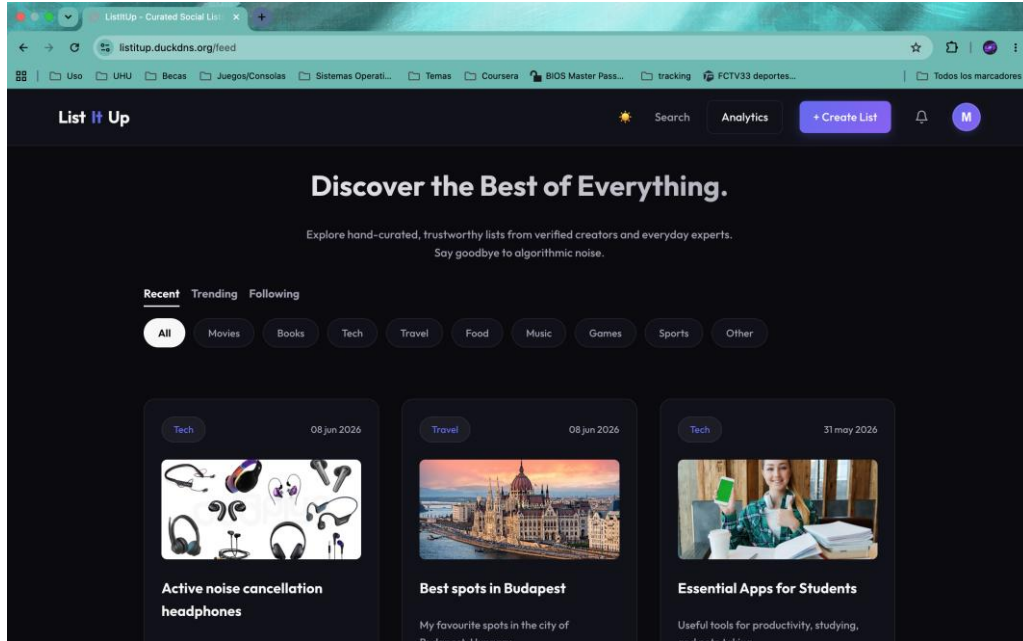
- *Type:* Usability
- *Description:* At least 80% of pilot users must be able to complete the core create-and-publish-list task without any staff assistance, measured during moderated usability sessions.
- *Verification Method:* Moderated usability testing sessions were made by 5 pilot users. Participants were instructed to complete the core "create-and-publish-list" task. Sessions were observed in real-time to ensure no staff assistance or guidance was provided. Success was recorded as a binary pass/fail based on independent completion.
- *Test Results:* 5 out of 5 pilot users (100%) successfully completed the core task of creating and publishing a list without any staff assistance. This result exceeds the minimum requirement of an 80% completion rate. Task execution times and minor user feedback were documented during the observation.

NF6 - Cross-Browser Responsiveness

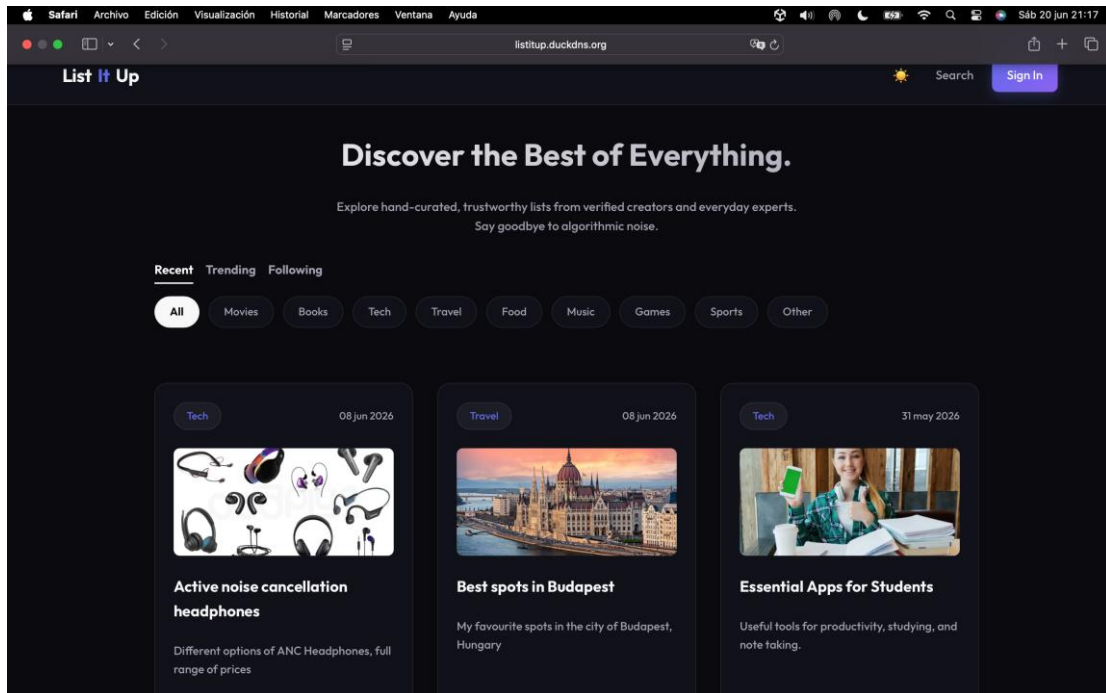
- *Type:* Compatibility / Responsiveness
- *Description:* The UI must render correctly and be fully functional across the two most recent stable versions of Chrome, Firefox, and Safari on both desktop and mobile viewports.
- *Verification Method:* Manual execution of core user flows across the two latest stable versions of Chrome, Firefox, and Safari on both desktop and mobile viewports. Visual and functional verification was documented using screenshots.
- *Test Results:* The UI renders correctly and is fully functional with no layout breaks across all specified browsers and devices.

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Google Chrome

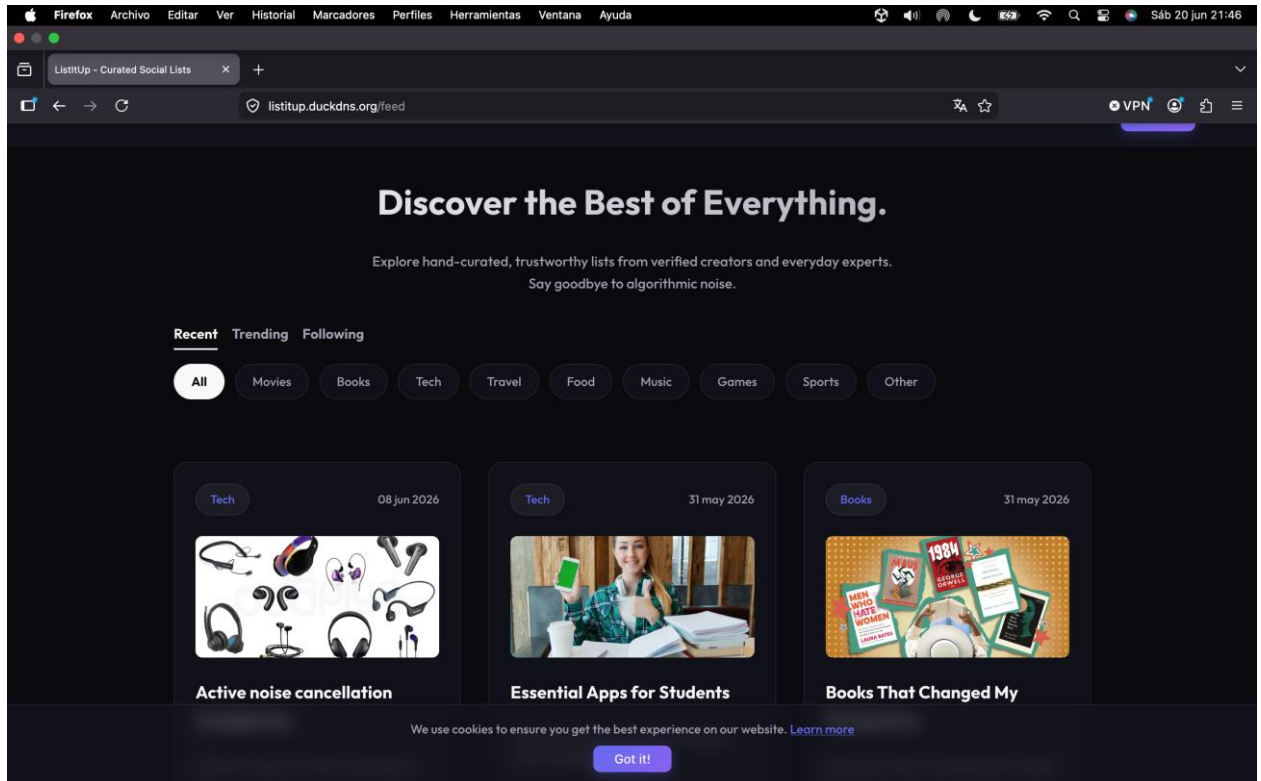


Safari

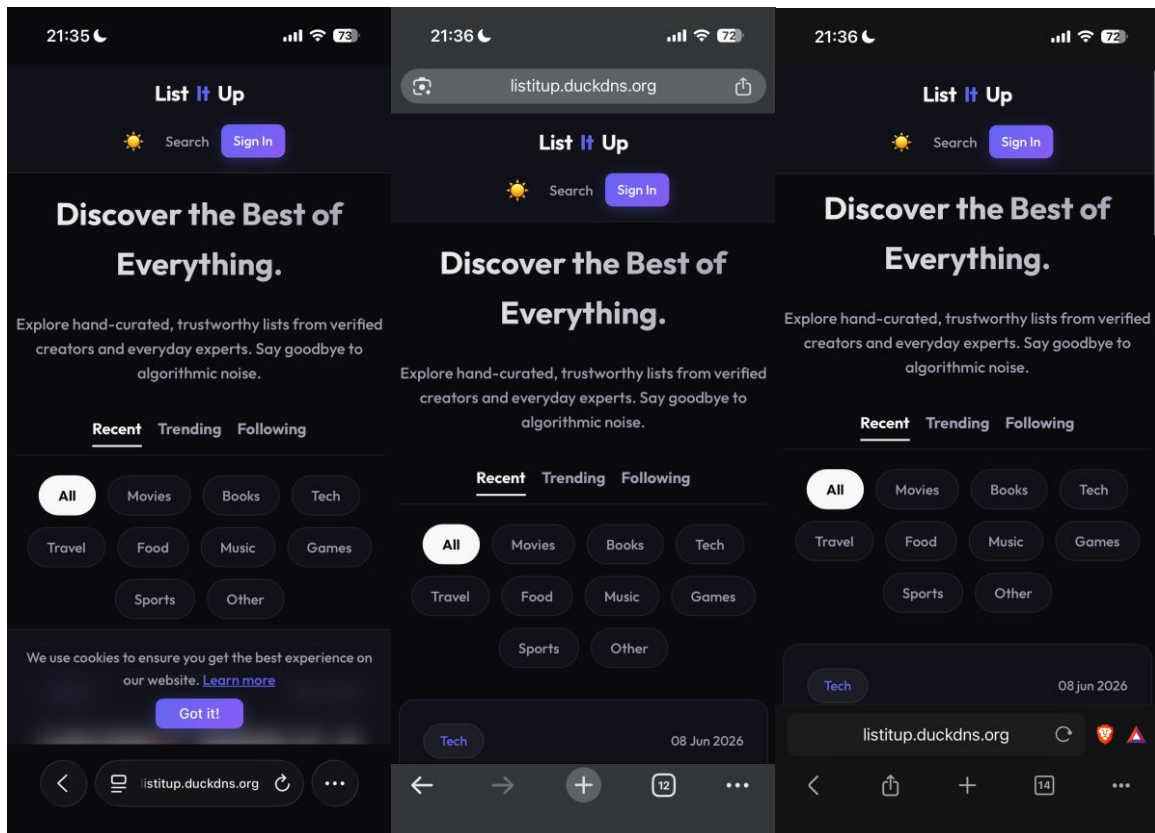


SSD 2026 - Project Report S4

Firefox



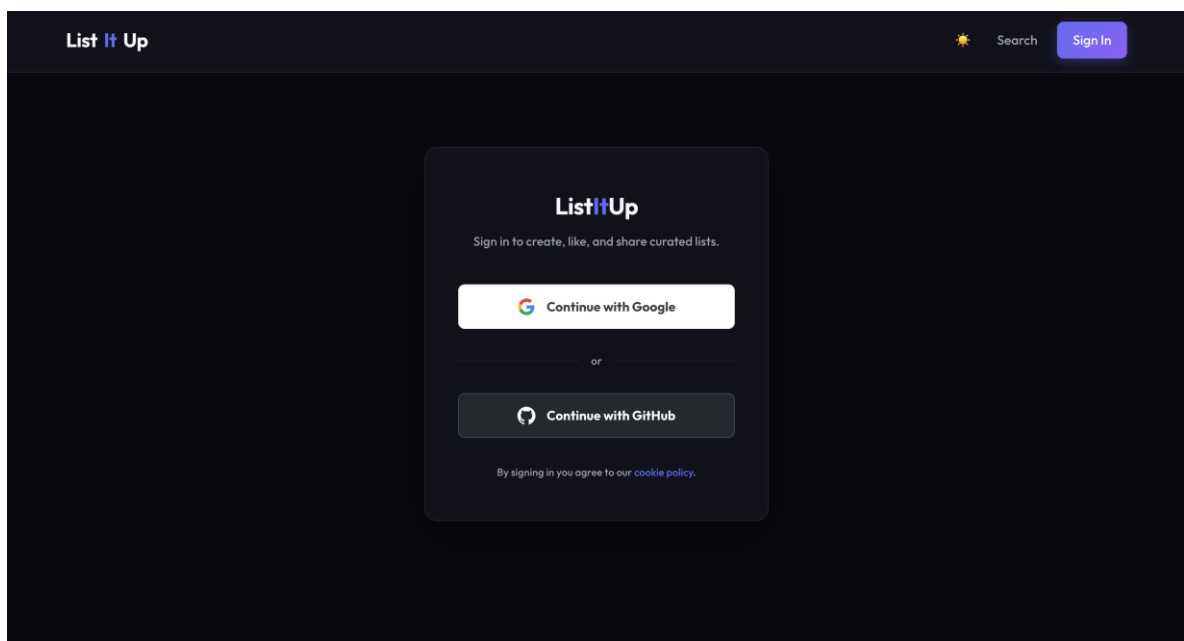
Mobile phone (Safari, Chrome and Brave)



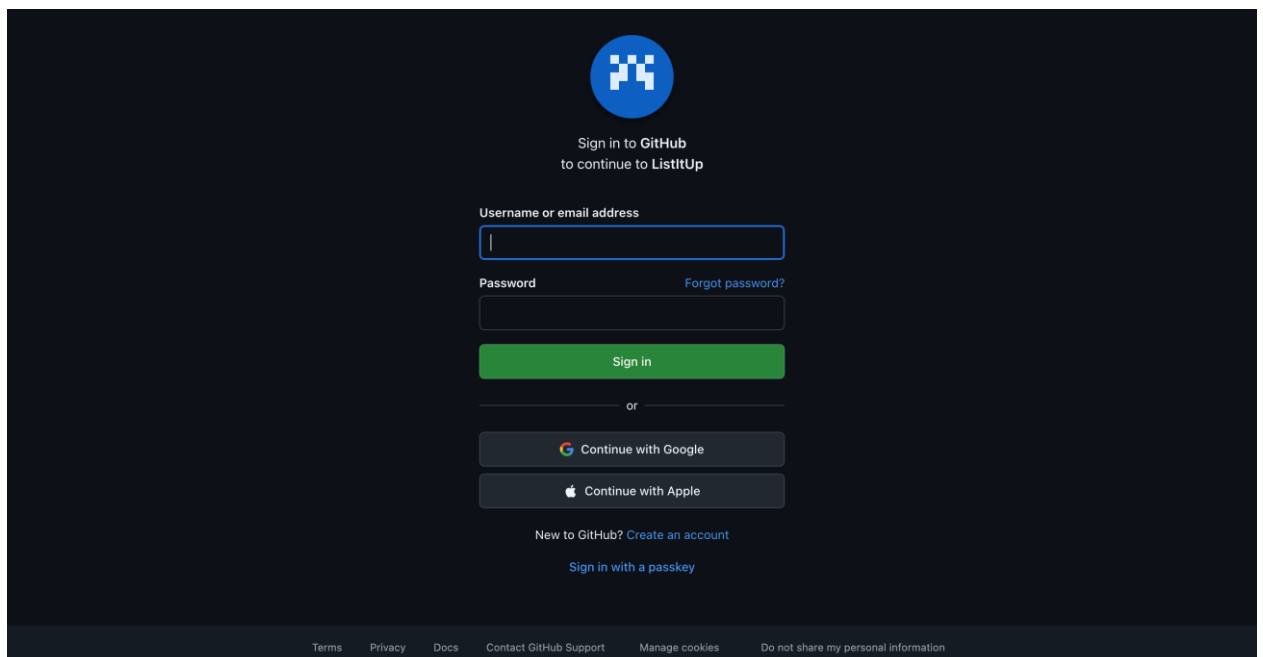
NF7 - OAuth 2.0 Social Login

- *Type:*
- Security / Authentication
- *Description:* The platform must support user authentication exclusively via OAuth 2.0 (Google and GitHub). No platform-specific passwords are stored or managed.
- *Verification Method:* Manual execution of the complete OAuth 2.0 authentication flow for both Google and GitHub providers. Verification includes confirming successful redirection, user consent handling, and session creation upon callback. Visual inspection confirmed the absence of any local password-based registration forms.
- *Test Results:* Users can successfully authenticate using both Google and GitHub OAuth providers.

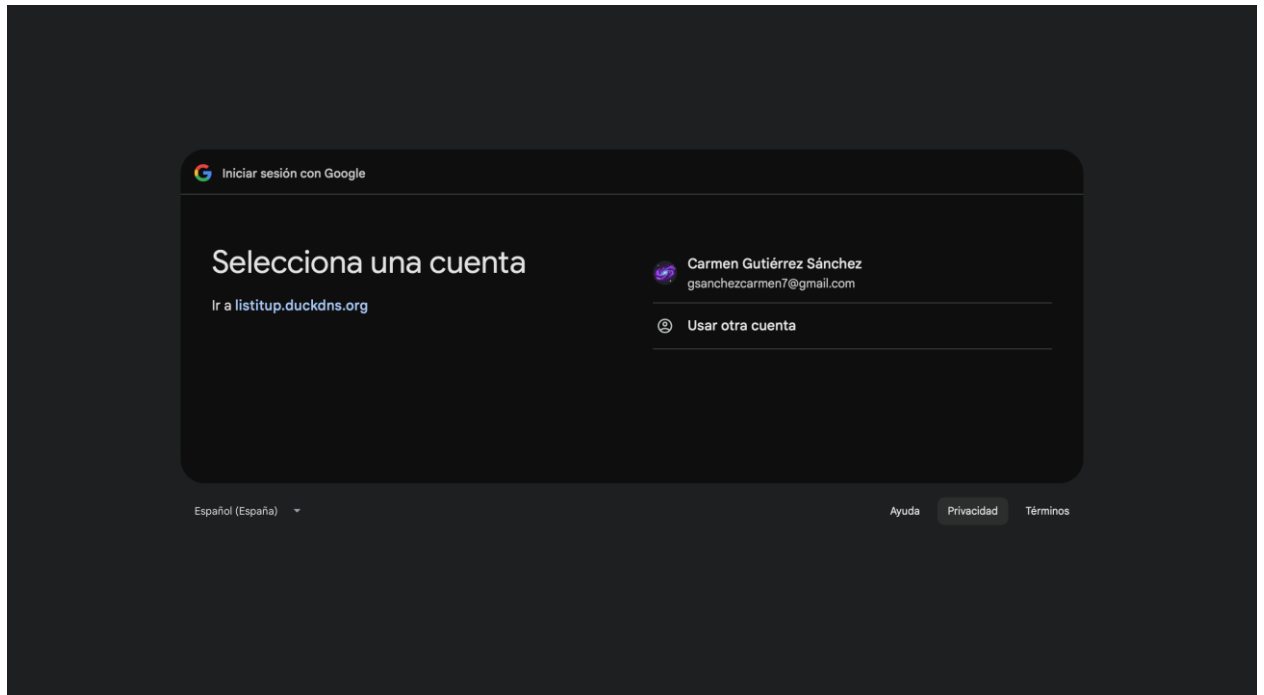
Login Panel



GitHub Option Panel



Google Option Panel



NF8 – User Profile Management

- *Type:* Usability
- Security / Authentication
- *Description:* Each user must have a publicly visible profile containing a username, biography, and profile picture. Verified creators must additionally display a verification badge on their profile.
- *Verification Method:* Inspect a registered user profile: verify that username, biography and profile picture fields are displayed. Log in as a verified creator and confirm that the verification badge appears on the profile page.
- *Test Results:* To verify the user profile management requirements, a manual inspection of the platform's public profile views was performed.
 - **Standard Profile Fields:** Reviewing a registered user's profile page confirmed that the system correctly renders the required public attributes: username, biography, and profile picture.
 - **Verification Badge:** Inspection of a profile belonging to a user with verified status confirmed that the dedicated verification badge is dynamically displayed on the page interface, visually distinguishing verified creators from standard accounts.

Profile Info Display:

List  Up

 Search

+ Create List





miryam

Role: STANDARD

0 followers | 1 following

Love books and club penguin

Profile picture updated successfully!

Edit Profile

Username

miryam

Save

Bio

Love books and club penguin

Save Bio

Danger Zone

Once you delete your account, there is no going back. Please be certain.

Delete Account

My Created Lists

My Saved Lists

My Liked Lists

You haven't created any lists yet.

© 2026 ListUp — Curated lists, shared with the world.

Verified Profile:

List  Up

 Search

+ Create List





Mamawevo

VERIFIED

2 followers | 1 following

Following

Joined June 2026

Mamawevo's Lists

4.2.2. Remaining Tests

We did a suite of automated unit and integration tests that have been implemented to ensure the robustness of the system's business logic, data access, and web layers.

As previously detailed in [Section 4.1.3 \(D. Test Artifacts\)](#), a total of 96 automated tests were developed. These tests are categorized as follows:

- **Unit Tests (Service Layer):** Isolated tests for UserService, ListService, ItemService, SocialService and AnalyticsService using standard mocking frameworks to validate business rules and calculations independently of the database.
- **Integration Tests (Web/Controller Layer):** Executed using MockMvc for ListController and AdminController to verify HTTP endpoints, security guards (OAuth2), and role-based access control (STANDARD vs. ADMIN).
- **Data Access Tests (Repository Layer):** Executed using @DataJpaTest for UserRepository and CuratedListRepository to validate custom database queries, pagination, and data persistence directly against a test database context.